Novell Pharmaceutical Laboratories

# Optimizing the Use of Somatostatin: From Acute Esophageal Variceal Hemorrhagic/Bleeding to Pancreatitis

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# 01

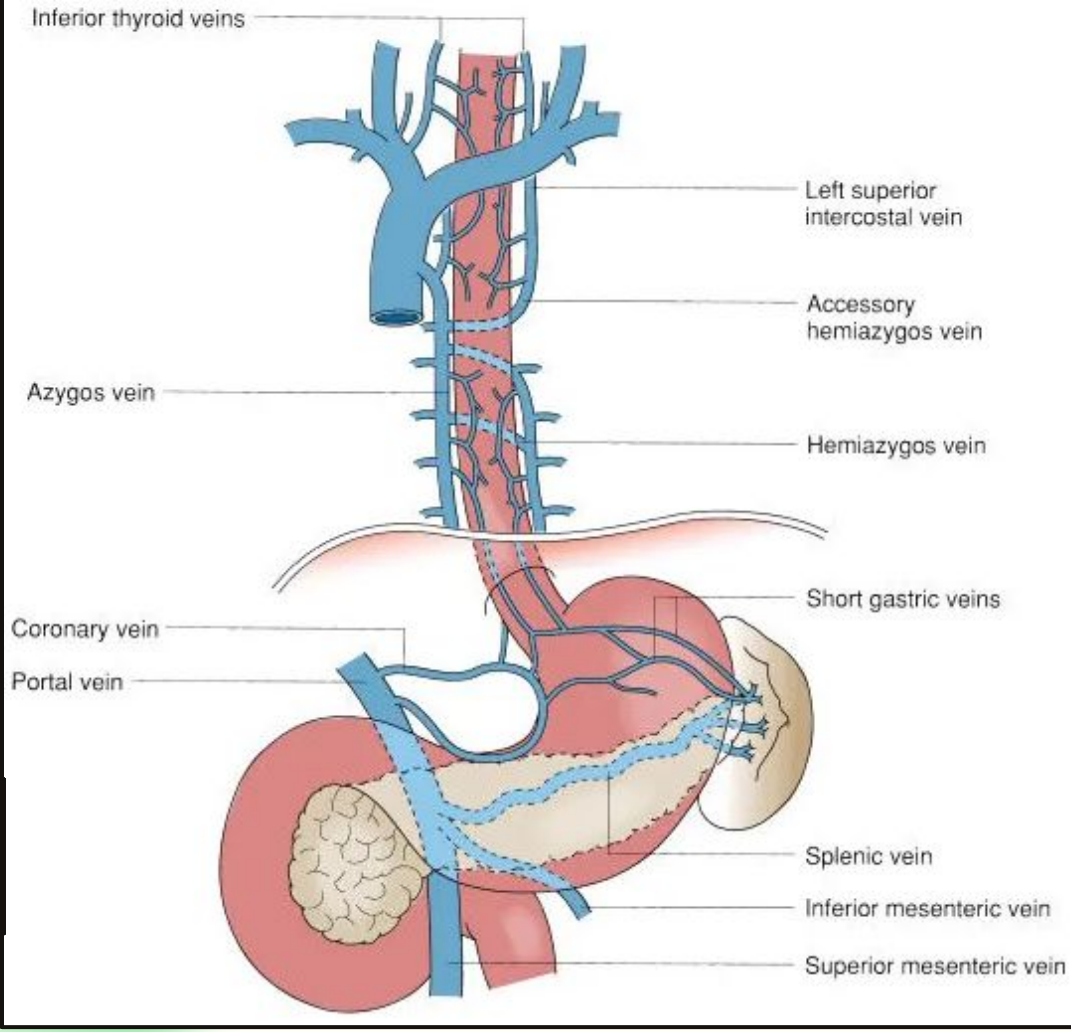
# Esophageal Varices

# Anatomi Vena Porta & Esofagus

Vena porta terbentuk dari mesenterika sup. & splenika

Tekanan portal fisiologis 5–10 mmHg

Vena submukosa esofagus bermuara ke azygos (aliran hepatofugal)



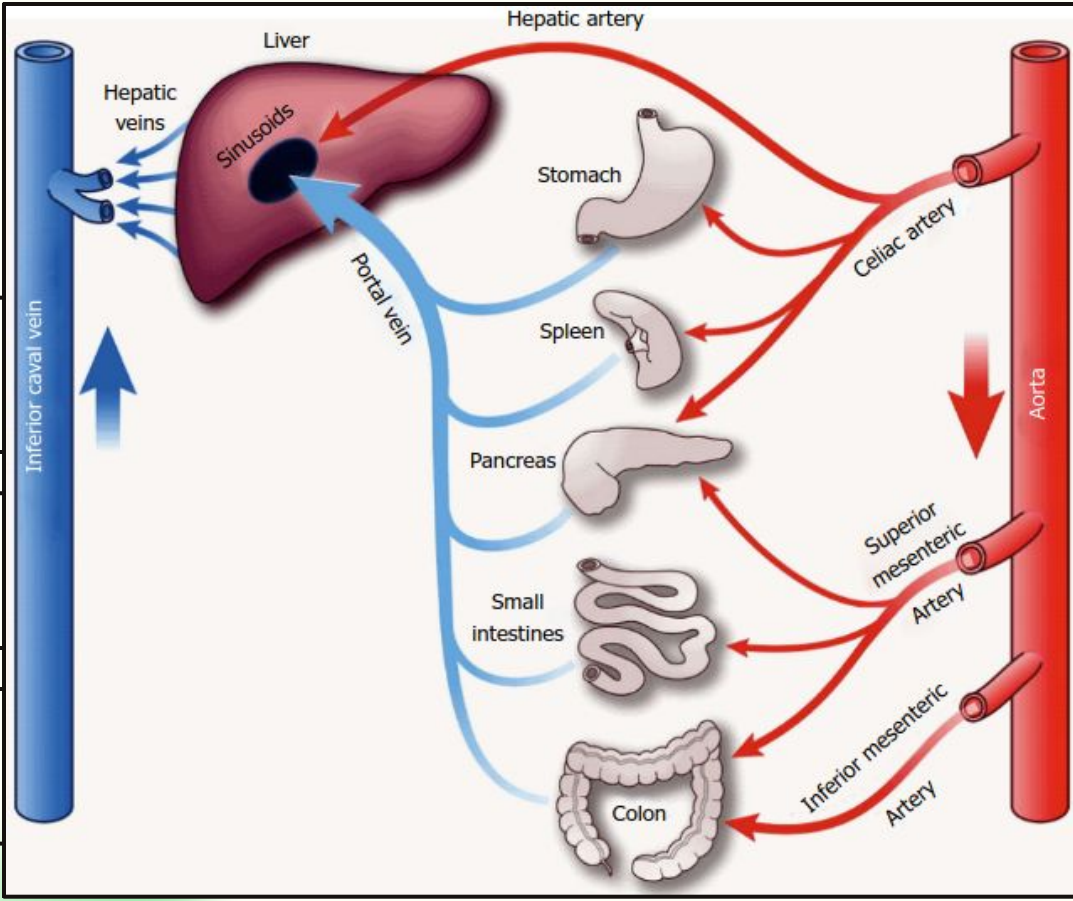

Sharma M, Rameshbabu CS. Collateral pathways in portal hypertension. J Clin Exp Hepatol. 2012;2(4):338-352. doi:10.1016/j.jceh.2012.08.001
Gambar diakses dari: https://basicmedicalkey.com/esophageal-anatomy-and-physiology-and-gastroesophageal-reflux-disease/

# Anatomi Sirkulasi Splanknik

Sirkulasi splanknik mencakup pembuluh gaster, usus, kolon, pankreas, limpa

Sirkulasi splanknik disuplai arteri celiaka, mesenterika superior & inferior

Menerima ±25 % curah jantung saat istirahat dan meningkat setelah makan




Prin M, Bakker J, Wagener G. Hepatosplanchnic circulation in cirrhosis and sepsis. World J Gastroenterol. 2015;21(9):2582-2592. doi:10.3748/wjg.v21.i9.2582

# Varises Esofagus

Dilatasi vena submukosa akibat **hipertensi portal**

Risiko pecah ↑ seiring diameter & tekanan → **Esophageal Variceal Hemorrhage / Bleeding of Variceal Esophagus (BEV)**

**15–25 %**

Mortalitas perdarahan 6-minggu

# Korelasi Hipertensi Portal & Varises Esofagus - Variceal Bleeding

**Table 1.** Correlation of portal pressure and clinical end-points in patients with liver cirrhosis

| Portal pressure | Clinical end-points |
|---|---|
| <5 mmHg | Normal |
| 5–10 mmHg | Mild portal hypertension |
| >10 mmHg | Clinically significant portal hypertension |
| >10 mmHg | Esophageal varices development, ascites, decompensation, hepatocellular carcinoma occurrence |
| >12 mmHg | Variceal bleeding |
| >16 mmHg | High mortality |
| >20 mmHg | Early rebleeding or failure to control bleeding |

Lesmana CRA, et al. Managing liver cirrhotic complications: Overview of esophageal and gastric varices. Clin Mol Hepatol 2020;26:444-460. https://doi.org/10.3350/cmh.2020.0022

# Epidemiologi Global

| 30–40% | 85% | 10–15% |
|---|---|---|
| Prevalensi varises esofagus: 30–40 % sirosis kompensasi | 85 % sirosis dekompensasi memiliki varises | Risiko perdarahan tahunan 10–15 % |

Elghezewi A, Hammad M, El-Dallal M, Mohamed M, Sherif A, Frandah W. Trends in Hospitalizations of Esophageal Varices From 2011 to 2018: A United States Nationwide Study. *Gastroenterology Res.* 2023;16(3):171-183. doi:10.14740/gr1627

# Epidemiologi Indonesia

**61,6%**

RSCM 2016-2017: 61,6 % varises esofagus

**51,8%**

51,8 % Etiologi utama: Hepatitis B

**73%**

Pasien laki-laki 73 %

**34%**

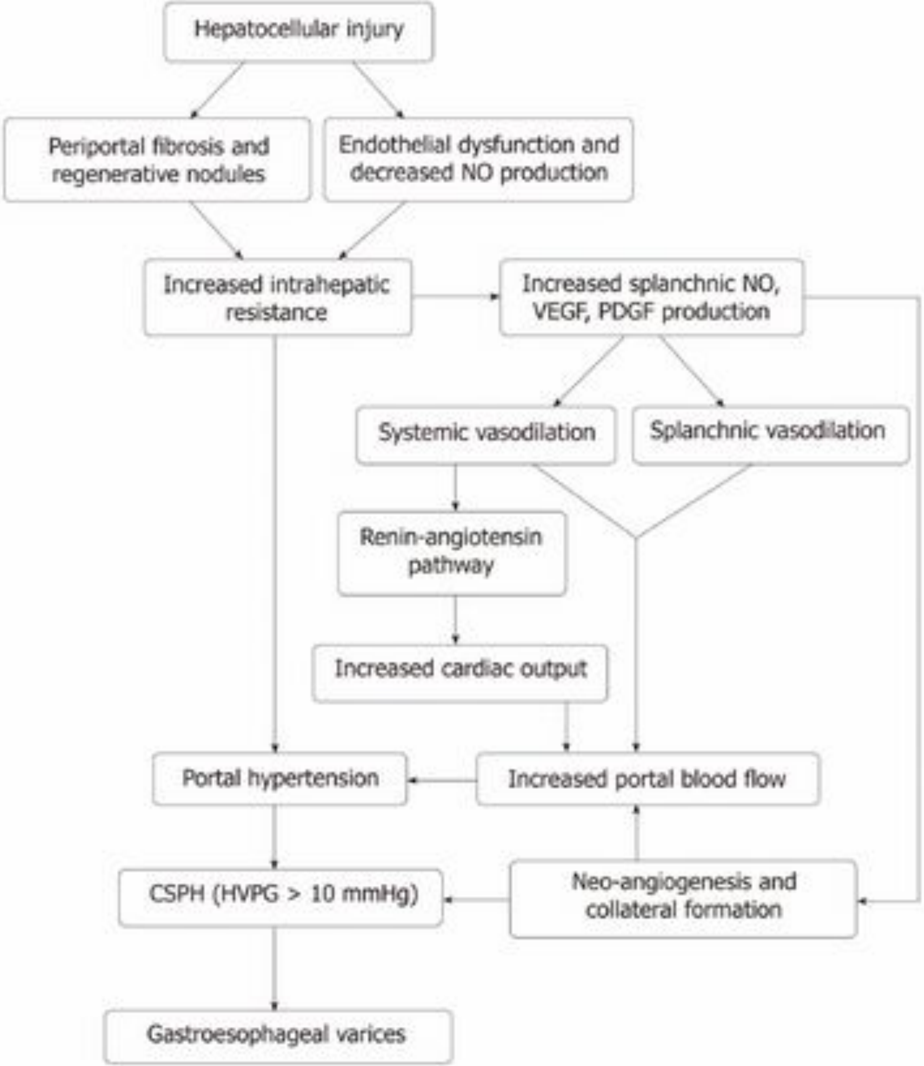
Usia >60 th 34 %

Kalista KF, et al. rofil Klinis Pasien Sirosis Hati dengan Varises Esofagus yang Menjalani Ligasi Varises Esofagus di Rumah Sakit Dr. Cipto Mangunkusumo. urnal Penyakit Dalam Indonesia. 2019;6(1):36-41.

# Etiologi Varises

| Sirosis | | | | Non–Sirosis | | |
|---|---|---|---|---|---|---|
| Sirosis alkohol | HBV | HCV | NAFLD/NASH meningkat global | Trombus vena porta | Schistosomiasis → fibrosis portal | Budd-Chiari |


Lesmana CRA, et al. Managing liver cirrhotic complications: Overview of esophageal and gastric varices. Clin Mol Hepatol 2020;26:444-460. https://doi.org/10.3350/cmh.2020.0022
Maruyama H, Yokosuka O. Pathophysiology of portal hypertension and esophageal varices. Int J Hepatol. 2012;2012:895787. doi:10.1155/2012/895787
Daniella, D., & Kristianto, M. O. (2018). Diagnosis dan Tatalaksana Non-Cirrhotic Portal Hypertension. Cermin Dunia Kedokteran, 45(12), 893–895. https://doi.org/10.55175/cdk.v45i12.680




# Patofisiologi

Fibrosis ↑ resistensi intrahepatik

↓

Vasodilatasi splanknik ↑ aliran portal

↓

HVPG ≥ 10 mmHg varises

↓

≥12 mmHg bleeding

↓

Angiogenesis kolateral via VEGF

LaBrecque D, Khan AG, Sarin SK, Mair AWL, dkk. Pedoman praktek WGO varises esofagus. Pedoman Global Organisasi Gastroenterologi Dunia. 2014. Diakses dari: https://www.worldgastroenterology.org/UserFiles/file/guidelines/esophageal-varices-english-2014.pdf

# Hepatic Venous Pressure Gradient (HVPG)

**No varices**

- *HVPG normal/<10 mmHg*

▼

**Small varices - No hemorrhage**

- *HVPG ≥10 mmHg*
- Varices development rate 8% per year

▼

**Large varices - No hemorrhage**

- *Hyper dynamic circulation*
- Progression from small to large 8% per year[2]

▼

**Variceal hemorrhage**

- *Pressure >variceal wall tension*
- Esophageal hemorrhage 5-15% per year
- Bleeding in patients with gastric varices is reported in approximately 25% in 2 years (higher for IGV1 and GOV2)[3].

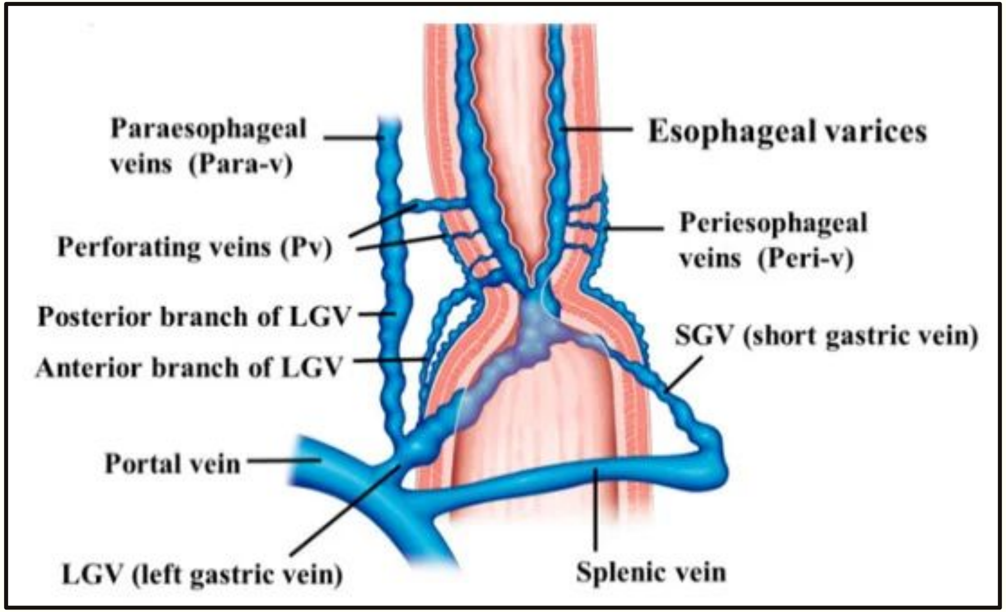
▼

**Recurrent hemorrhage**

- *Persistence of portal pressure and variceal status*

LaBrecque D, Khan AG, Sarin SK, Mair AWL, dkk. Pedoman praktek WGO varises esofagus. Pedoman Global Organisasi Gastroenterologi Dunia. 2014. Diakses dari: https://www.worldgastroenterology.org/UserFiles/file/guidelines/esophageal-varices-english-2014.pdf

# Gambaran Varises Esofagus




Pembuluh darah yang terlibat: Portal Vein, LGB, SGV, Peri-V, Para-V

Nagashima K, Irisawa A, Tominaga K, et al. The Role of Endoscopic Ultrasound for Esophageal Varices. Diagnostics (Basel). 2020;10(12):1007. Published 2020 Nov 25. doi:10.3390/diagnostics10121007

# Guideline for Diagnosing Esophageal Varices

| | | | |
|---|---|---|---|
| **1** | **A screening esophagogastroduodenoscopy (EGD) for the diagnosis of esophageal and gastric varices is recommended when a diagnosis of cirrhosis has been made** | | |
| **2** | **Surveillance endoscopies are recommended on the basis of the level of cirrhosis and the presence and size of the varices:** | | |
| | *Patients with* | *and* | *Repeat EGD* |
| | Compensated cirrhosis | No varices | Every 2–3 years |
| | | Small varices | Every 1–2 years |
| | Decompensated cirrhosis | | Yearly intervals |
| **3** | **Progression of gastrointestinal varices can be determined on the basis of the size classification at the time of EGD. In practice, the recommendations for medium-sized varices in the three-size classification are the same as for large varices in the two-size classification:** | | |
| | *Size of varix* | *Two-size classification* | *Three-size classification* |
| | Small | < 5 mm | Minimally elevated veins above the esophageal mucosal surface |
| | Medium | – | Tortuous veins occupying less than one-third of the esophageal lumen |
| | Large | > 5 mm | Occupying more than one-third of the esophageal lumen |
| **4** | **Variceal hemorrhage is diagnosed on the basis of one of the following findings on endoscopy:** | | |
| | • Active bleeding from a varix<br>• "White nipple" overlying a varix<br>• Clots overlying a varix<br>• Varices with no other potential source of bleeding | | |

LaBrecque D, Khan AG, Sarin SK, Mair AWL, dkk. Pedoman praktek WGO varises esofagus. Pedoman Global Organisasi Gastroenterologi Dunia. 2014. Diakses dari: https://www.worldgastroenterology.org/UserFiles/file/guidelines/esophageal-varices-english-2014.pdf

# Pemeriksaan Non-invasif & Invasif

| Metode Diagnostik | Temuan Pemeriksaan |
|---|---|
| **Non-invasif** | |
| **Ultrasonografi (USG)** | |
| Hepar | permukaan ireguler, inhomogen, lesi fokal pada hepar |
| Vena porta | dilatasi, trombosis +/- |
| Limpa | splenomegali |
| Kolateral portosistemik, asites | + |
| Pemeriksaan CEUS | penyangatan periportal lambat/heterogen/homogen |
| Pencitraan potong lintang | karakterisasi lesi fokal hepar yang lebih baik |
| **Elastografi** | |
| Kekakuan hati | ↑ |
| Kekakuan limpa | ↑ |
| **Invasif** | |
| **Biopsi hati** | gambaran fibrosis dan perubahan arsitektur hepar |
| **Hemodinamik hati** | FHVP normal, WHVP↑, HVPG↑, keadaan sirkulasi hiperdinamik |
| **Endoskopi** | varises esofagus dan gastropati hipertensi portal lebih sering, varises gaster jarang ditemukan |

Konsensus Nasional: Penatalaksanaan hipertensi portal di Indonesia. Perhimpunan Peneliti Hati Indonesia. 2021

# WGO Recommendations for First-Line Management

**No varices**
Repeat Endoscopy in 2-3 years

▼

**Small varices - No hemorrhage**
Repeat Endoscopy in 1-2 years

▼

**Medium/large varices - No hemorrhage**
β-blockers (propranolol, nadolol, or carvedilol)
EVL if β-blockers are not tolerated

▼

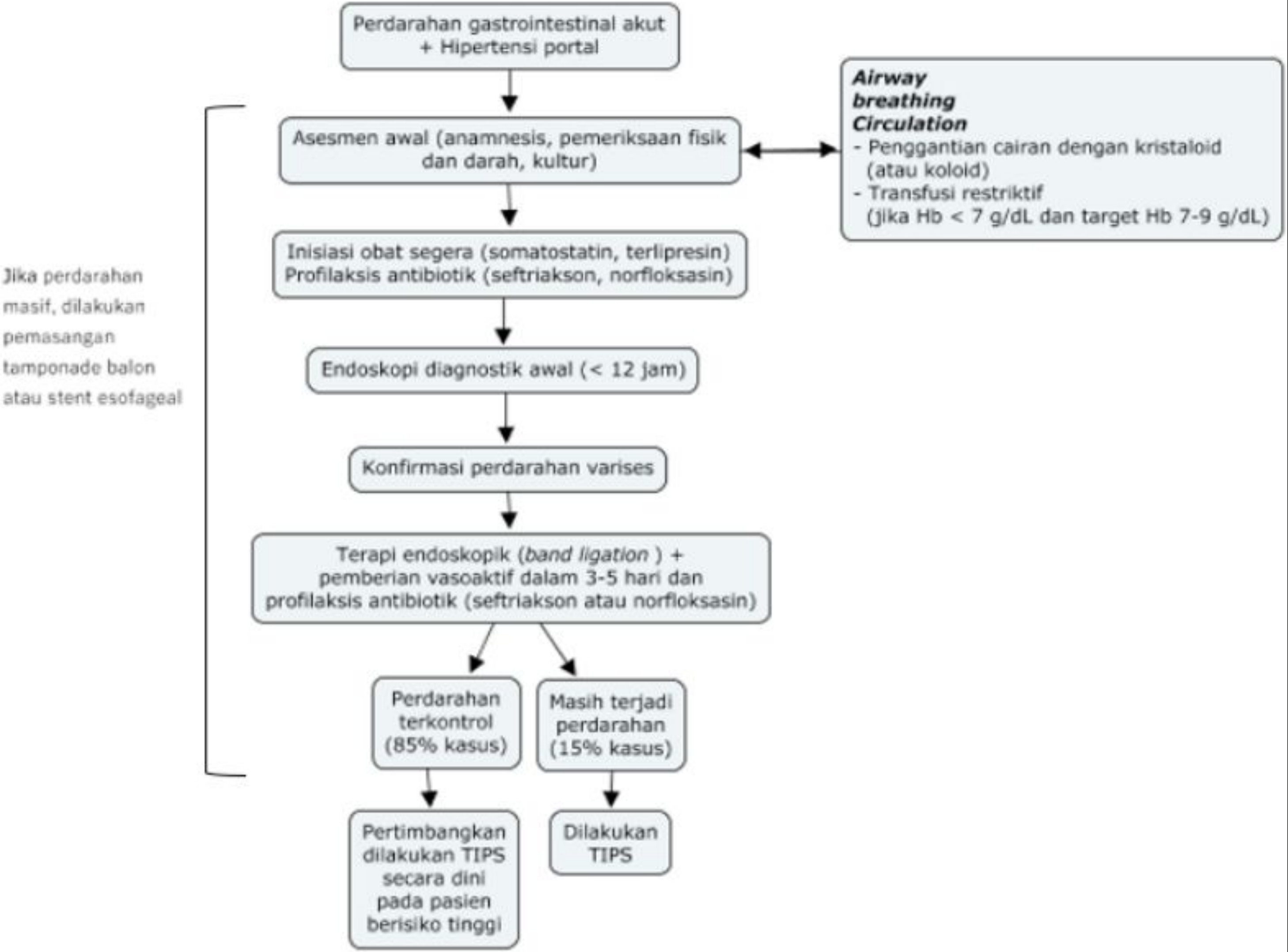
**Variceal hemorrhage**
Specific therapy: safe vasoactive drug + EVL

▼

**Recurrent hemorrhage**
β-blockers +/- ISMN or EVL
β-blockers + EVL

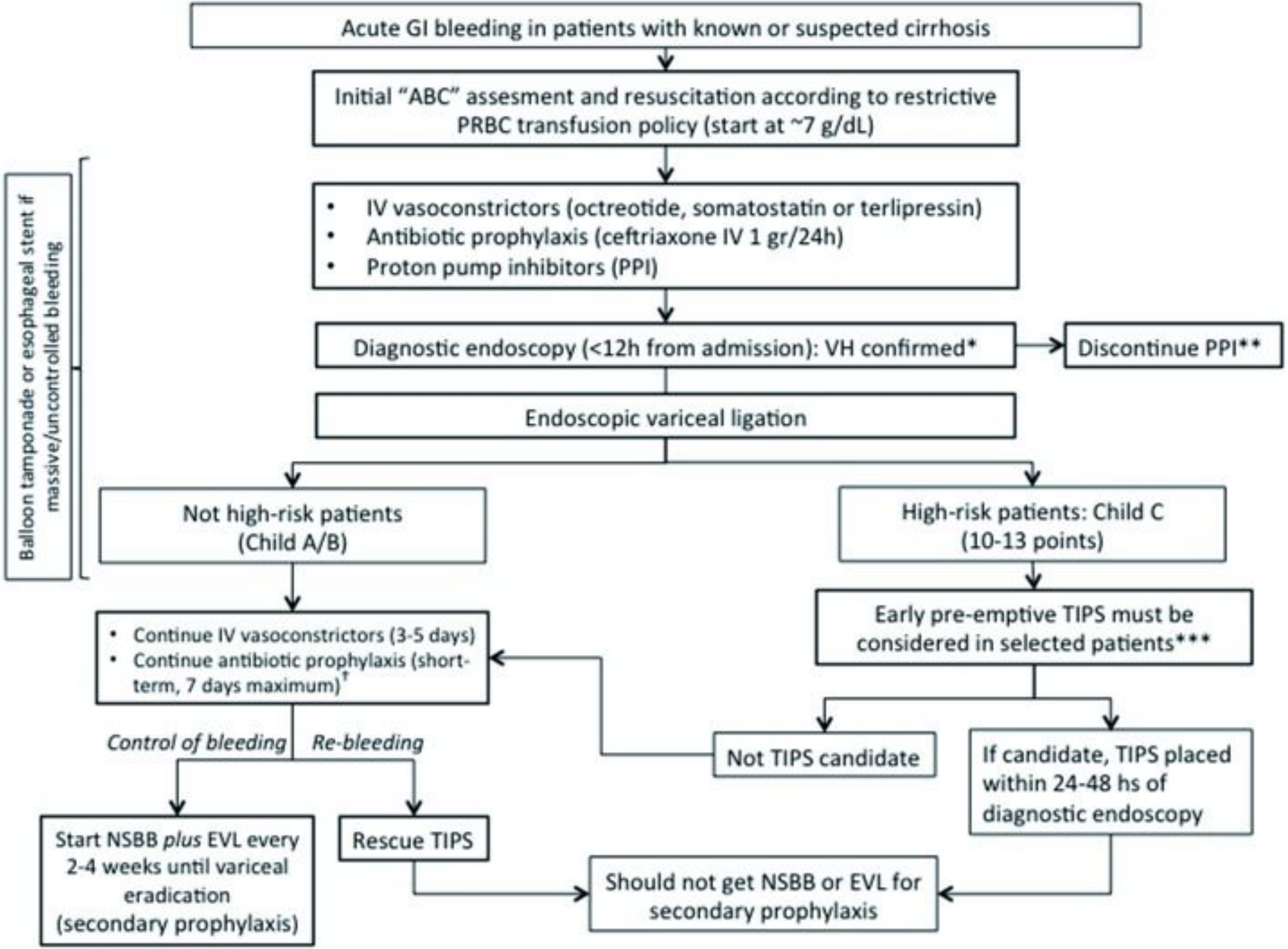
EVL: endoscopic variceal ligation
ISMN: isosorbide 5–mononitrate

LaBrecque D, Khan AG, Sarin SK, Mair AWL, dkk. Pedoman praktek WGO varises esofagus. Pedoman Global Organisasi Gastroenterologi Dunia. 2014. Diakses dari: https://www.worldgastroenterology.org/UserFiles/file/guidelines/esophageal-varices-english-2014.pdf




# Konsensus Nasional Perhimpunan Peneliti Hati Indonesia

Konsensus Nasional: Penatalaksanaan hipertensi portal di Indonesia. Perhimpunan Peneliti Hati Indonesia. 2021




# Management of Acute Variceal Hemorrhage

Zanetto A, Garcia-Tsao G. Management of acute variceal hemorrhage. F1000Res. 2019;8:F1000 Faculty Rev-966. Published 2019 Jun 25. doi:10.12688/f1000research.18807.1

# AGA Recommendations Vasoactive Drugs

**Table 1.** Standard Regimens for the Use of Vasoactive Drugs in the Management of Acute Variceal Hemorrhage and Prevention of Early Rebleeding

| Drug | Standard regimen | Duration, *d* |
|---|---|---|
| Octreotide (somatostatin analogue) | 50 μg IV bolus, followed by continuous IV infusion at 50 μg/h; additional IV boluses can be given in case of ongoing bleeding | 2–5 |
| Somatostatin | 250 μg IV bolus, followed by continuous IV infusion at 250–500 μg/h; additional IV boluses can be given in case of ongoing bleeding | 2–5 |
| Terlipressin (vasopressin analogue)[a] | Initial 48 h: 2 mg IV every 4 h until bleeding is controlled<br>Maintenance: 1 mg IV every 4 h | 2–5 |

[a]Although terlipressin has been evaluated for AVH and is one of the standard regimens outside of the United States, the FDA label does not include this indication.

Garcia-Tsao G, Abraldes JG, Rich NE, Wong VWS. AGA Clinical Practice Update on the Use of Vasoactive Drugs and Intravenous Albumin in Cirrhosis: Expert Review. Gastroenterology 2024;166:202–210. https://doi.org/10.1053/j.gastro.2023.10.016

# AASLD Recommendations Vasoactive Drugs

AASLD Practice Guidance on risk stratification and management of portal hypertension and varices in cirrhosis. 2023. DOI: 10.1097/HEP.0000000000000647

**TABLE 5** Vasoactive agents for acute variceal bleeding.

| Agent | Dosing | Duration |
|---|---|---|
| Octreotide | Initial i.v. bolus of 50 mcg and continue infusion at a rate of 25–50 mcg/hour[151–153] | 2–5 d |
| Somatostatin | Initial i.v. bolus of 250 mcg and continue infusion at a rate of 250–500 mcg/hour[154,155] | 2–5 d |
| Terlipressin[a] | Initial 24–48 hours: 2 mg i.v. every 4–6 hours and then 1 mg i.v. every 4–6 hours [154,156–158] | 2–5 d |

[a]Not approved for this indication in North America.

References: Garcia-Tsao et al. *Hepatology*. January 2017[1]; Seo et al. *Hepatology*. September 2014.[159]

25. All patients with known or suspected cirrhosis presenting with acute gastrointestinal bleeding should be initiated on vasoactive therapy (e.g., somatostatin, octreotide or terlipressin if available; see Table 5) and intravenous antibacterial therapy as soon as possible.

# 02

# Pankreatitis

# Definition

Pankreatitis adalah proses inflamasi pada pankreas yang dapat terjadi secara akut maupun kronis


PERKUMPULAN GASTROENTEROLOGI INDONESIA

(PGI)

KONSENSUS NASIONAL PENATALAKSANAAN PANKREATITIS AKUT DI INDONESIA

Manan HC, Makmun HD, Aulia C. Konsensus nasional penatalaksanaan pankreatitis akut di Indonesia. Perkumpulan Gastroenterologi Indonesia (PGI). 2011. https://pbpgigastro.com/wp-content/uploads/2024/07/Konsensus-Nasional-Penatalaksanaa-Pankreatitis-Akut-di-Indonesia-Tahun-2011.pdf
Pratama H. Tatalaksana Pankreatitis Akut. CDK-238. 2016;43(3):190-4. https://media.neliti.com/media/publications/401267-tatalaksana-pankreatitis-akut-3c2dd11c.pdf

| Peneliti (tahun) | Negara | Subjek | Insidens (per 100.000/tahun) |
|---|---|---|---|
| Banks (2002) | Inggris, Belanda | Serangan pertama/ rekurensi | 5-10 |
| | Skotlandia, Denmark | | 25-35 |
| | USA, Finlandia | | 70-80 |
| Tinto, *et al.* (2002) | UK | Serangan pertama/ rekurensi | 14,5 – 20,7 |
| Andersson, *et al.* (2004) | Swedia | Serangan pertama/ rekurensi | 30 |
| Lankisch, *et al.* (2002) | Jerman | Serangan pertama/ rekurensi | 19,7 |
| Gislason, *et al.* (2004) | Norwegia | Serangan pertama/ rekurensi | 30,6 |
| | | Serangan pertama | 20 |
| Birgisson, *et al.* (2002) | Islandia | Serangan pertama | 32 |
| Floyd, *et al.* (2002) | Denmark | Pria | 27,1 |
| | | Wanita | 37,8 |
| Survey nasional Jepang | Jepang | Serangan pertama/ rekurensi | 12,1 |
| Survey nasional Jepang | Jepang | Serangan pertama/ rekurensi (total) | 15,4 |
| | | Serangan pertama/ rekurensi (pria) | 20,5 |
| | | Serangan pertama/ rekurensi (wanita) | 10,6 |

# Epidemiology

Insidensi di Amerika Serikat mencapai 79,8 per 100.000 penduduk/tahun

# Etiologi

Di Indonesia etiologi utama sering bersifat idiopatik

30–50 %

Batu Empedu: Merupakan penyebab terbanyak (30–50% kasus)
Terutama akibat obstruksi duktus biliaris komunis

20–30 %

Alkohol: Menyumbang 20–30% kasus

15–25 %

Alkohol: Menyumbang 20–30% kasus

Manan HC, Makmun HD, Aulia C. Konsensus nasional penatalaksanaan pankreatitis akut di Indonesia. Perkumpulan Gastroenterologi Indonesia (PGI). 2011. https://pbpgigastro.com/wp-content/uploads/2024/07/Konsensus-Nasional-Penatalaksanaa-Pankreatitis-Akut-di-Indonesia-Tahun-2011.pdf

# EtiologI berdasarkan Mekanisme

| | |
|---|---|
| Bilier:<br>Batu empedu, microlithiasis, atau sludge empedu | Toksik:<br>Paparan alkohol, sengatan kalajengking, insektisida organofosfat |
| Obstruksi Mekanik saluran pankreas<br>Tumor (ganas atau jinak), striktur, disfungsi sfingter Oddi | Obat-obatan:<br>Asosiasi definitif:<br>Sulfasalazin, mesalazin, asparaginase, pentamidine, azatioprine, dan didanosine<br>Asosiasi probable, possible:<br>Chlorthalidone, siklosporin, amiodarone, carbamazepine, dan lain-lain. |
| Infeksi:<br>Virus: mumps, Coxsackie A, HIV, dengue, hepatitis B, varicella-zoster.<br>Bakteri: Tuberkulosis, mycoplasma, leptospira, salmonella, legionella.<br>Parasit: Ascaris, Cryptosporidium, Toxoplasma | |
| Pasca Tindakan:<br>Prosedur ERCP atau tindakan bedah lain (abdomen/jantung) | Autoimun:<br>Lupus eritematosus sistemik, sindrom Sjögren, primary sclerosing cholangitis, hepatitis autoimun, serta penyakit celiac |
| Metabolik:<br>Hiperkalsemia, hipertrigliseridemia | Lain-lain:<br>Faktor genetik (familial), trauma, penyakit Crohn, kehamilan |

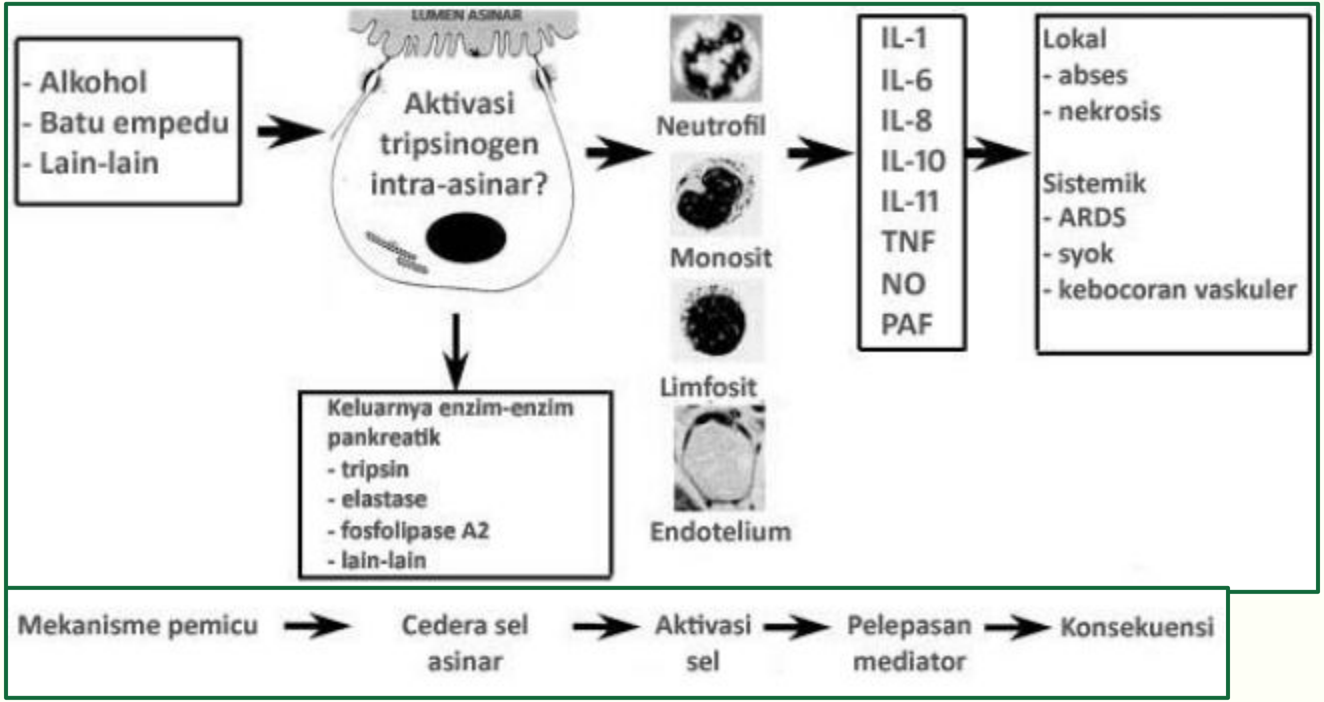
Manan HC, Makmun HD, Aulia C. Konsensus nasional penatalaksanaan pankreatitis akut di Indonesia. Perkumpulan Gastroenterologi Indonesia (PGI). 2011. https://pbpgigastro.com/wp-content/uploads/2024/07/Konsensus-Nasional-Penatalaksanaa-Pankreatitis-Akut-di-Indonesia-Tahun-2011.pdf

# Pathophysiology

| | |
|---|---|
| Fase Pertama | Aktivasi Prematur Tripsin<br>Terjadi di dalam sel-sel asinar pankreas.<br>Mekanisme utama adalah gangguan pensinyalan kalsium yang mengakibatkan aktivasi enzim tripsin secara prematur, sehingga enzim-enzim pencernaan teraktivasi di dalam pankreas sendiri yang semestinya hanya aktif pada lumen usus |
| Fase Kedua | Reaksi Peradangan Lokal<br>Cedera pada pankreas memicu reaksi inflamasi lokal<br>Berbagai mediator dan sitokin dilepaskan, yang kemudian mengaktivasi sel-sel radang dan merangsang mekanisme peradangan di tingkat mikrosirkulasi<br>Proses inflamasi ini memicu kerusakan jaringan lebih lanjut dan perkembangan tanda-tanda klinis peradangan |
| Fase Ketiga | Peradangan Ekstrapankreas<br>Inflamasi yang awalnya terjadi secara lokal menyebar ke jaringan sekitarnya dan mempengaruhi organ lain melalui sirkulasi sistemik |

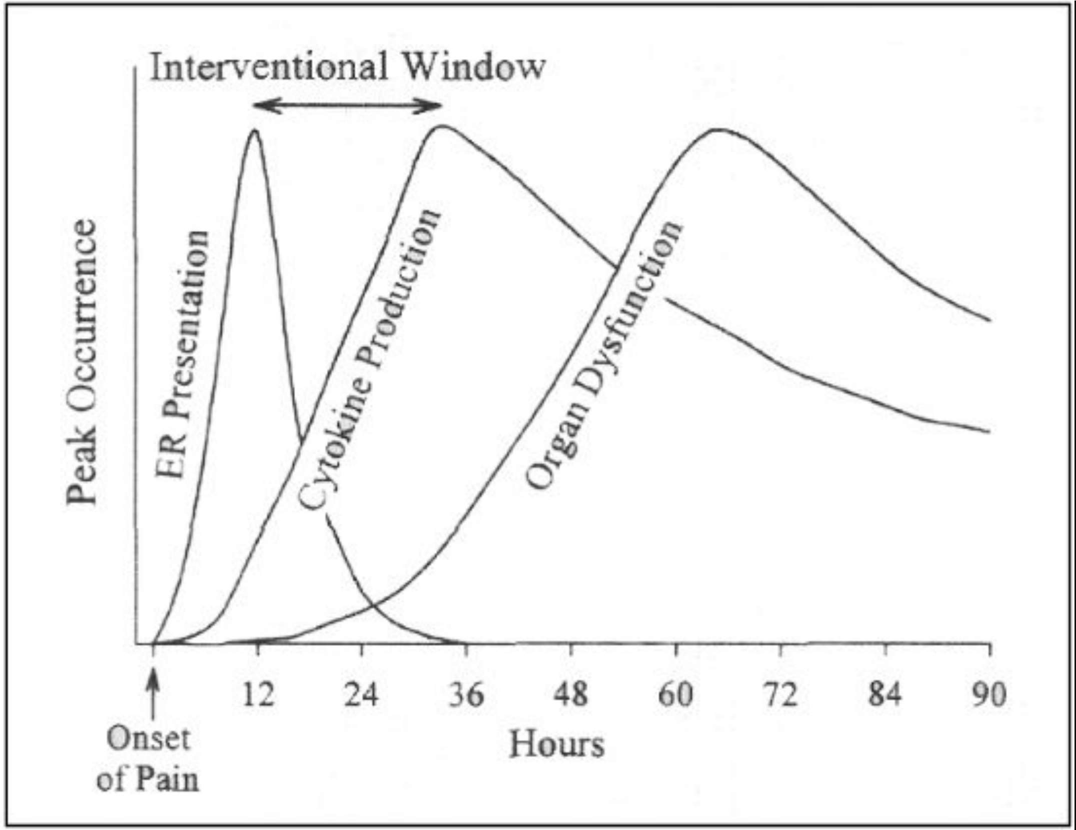
Manan HC, Makmun HD, Aulia C. Konsensus nasional penatalaksanaan pankreatitis akut di Indonesia. Perkumpulan Gastroenterologi Indonesia (PGI). 2011. https://pbpgigastro.com/wp-content/uploads/2024/07/Konsensus-Nasional-Penatalaksanaa-Pankreatitis-Akut-di-Indonesia-Tahun-2011.pdf

# Pathophysiology




Manan HC, Makmun HD, Aulia C. Konsensus nasional penatalaksanaan pankreatitis akut di Indonesia. Perkumpulan Gastroenterologi Indonesia (PGI). 2011. https://pbpgigastro.com/wp-content/uploads/2024/07/Konsensus-Nasional-Penatalaksanaa-Pankreatitis-Akut-di-Indonesia-Tahun-2011.pdf

# Perjalanan Penyakit Pankreatitis Akut




Umumnya pasien datang dalam 18 jam setelah munculnya nyeri.
Setelah itu, terdapat periode produksi sitokin proinflamasi yang berlangsung beberapa hari
Disfungsi organ dapat terjadi pada saat ini, sebagian besar penderita mengalami manifestasi sistemik yang berat pada 1–4 hari berikutnya
Hal tersebut menciptakan suatu jeda waktu untuk dapat melakukan intervensi terapeutik (interventional window)


Manan HC, Makmun HD, Aulia C. Konsensus nasional penatalaksanaan pankreatitis akut di Indonesia. Perkumpulan Gastroenterologi Indonesia (PGI). 2011. https://pbpgigastro.com/wp-content/uploads/2024/07/Konsensus-Nasional-Penatalaksanaa-Pankreatitis-Akut-di-Indonesia-Tahun-2011.pdf

# Diagnosis

Diagnosis ditegakkan apabila terdapat setidaknya 2 dari 3 komponen berikut:

- Nyeri Perut Khas: Umumnya terasa tajam, konstan, berlokasi di epigastrium atau kuadran atas (bisa menjalar ke punggung)
- Peningkatan Kadar Enzim: Kadar amilase dan/atau lipase serum meningkat lebih dari tiga kali batas atas normal
- Temuan Pencitraan: Temuan khas pada USG atau CT scan (dengan kontras) yang mendukung diagnosis

Temuan Klinis dan Pemeriksaan Fisik

- Nyeri abdomen akut sering disertai mual & muntah
- Pankreatitis ringan, hanya ditemukan nyeri tekan di daerah epigastrium
- Tanda–tanda khusus:
  - Cullen Sign (ekstravasasi eksudat ke area periumbilikus),
  - Grey–Turner (pembesaran pada area pinggang) dapat muncul pada kasus yang lebih berat

Manan HC, Makmun HD, Aulia C. Konsensus nasional penatalaksanaan pankreatitis akut di Indonesia. Perkumpulan Gastroenterologi Indonesia (PGI). 2011. https://pbpgigastro.com/wp-content/uploads/2024/07/Konsensus-Nasional-Penatalaksanaa-Pankreatitis-Akut-di-Indonesia-Tahun-2011.pdf

# Stratifikasi Risiko dan Pencitraan

| |
|---|
| Skor Prognostik:<br>Skor APACHE II, Ranson, Imrie, Glasgow, BISAP, dan HAPS digunakan untuk menilai tingkat keparahan |
| Pencitraan:<br>USG Abdomen: Deteksi batu empedu, pelebaran duktus biliaris<br>CT Scan dengan Kontras: 2–3 hari setelah masuk rumah sakit → membedakan antara pankreatitis interstisial dan nekrotikans,menilai derajat keparahan (kriteria Balthazar–Ranson)<br>Modalitas lain: MRI, MRCP, atau ERCP digunakan sesuai indikasi (mengevaluasi penyebab atau komplikasi) |

Manan HC, Makmun HD, Aulia C. Konsensus nasional penatalaksanaan pankreatitis akut di Indonesia. Perkumpulan Gastroenterologi Indonesia (PGI). 2011. https://pbpgigastro.com/wp-content/uploads/2024/07/Konsensus-Nasional-Penatalaksanaa-Pankreatitis-Akut-di-Indonesia-Tahun-2011.pdf

# Stratifikasi Risiko dan Pencitraan

**Tabel 1.** Skor APACHE II[24]

| | | +4 | +3 | +2 | +1 | 0 | +1 | +2 | +3 | +4 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Temperatur rektal (°C) | >41 | 39-40,9 | | 38-38,9 | 36-38.4 | 34-35.9 | 32-33.9 | 30-31.9 | <29.9 |
| 2 | Tekanan arterial rata-rata (mmHg) | >160 | 130-159 | 110-129 | | 70-109 | | 50-69 | | <49 |
| 3 | Denyut jantung (x/mnt) | >180 | 140-179 | 110-139 | | 70-109 | | 50-69 | 40-54 | <39 |
| 4 | Frekuensi napas (x/mnt) | >50 | 35-49 | | 25-34 | 12-24 | 10-11 | 6-9 | | <5 |
| 5 | Oxygen delivery (mL/mnt) | >500 | 350-499 | 200-349 | | <200 | | | | |
| 6 | $PO_2$ (mmHg) | | | | | >70 | 61-70 | | 55-60 | >55 |
| 7 | pH arterial | >7,7 | 7.6-7.69 | | 7.5-7.59 | 7.3-7.49 | | 7.25-7.3 | 7.15-7.2 | <7.15 |
| 8 | Natrium serum (mmol/L) | >180 | 160-179 | 155-159 | 150-154 | 130-149 | | 120-129 | 111-119 | <110 |
| 9 | Kalium serum (mmol/L) | >7 | 6-6.9 | | 5.5-5.9 | 3.5-5.4 | 3-3.4 | 2.5-2.9 | | <2.5 |
| 10 | Kreatinin serum (mg/dL) | >3,5 | 2-3.4 | 1.5-1.9 | | 0.6-1.4 | | <0.6 | | |
| 11 | Hematokrit (%) | >60 | | 50-59.9 | 46-49.9 | 30-45.9 | | 20-20.0 | | <20 |
| 12 | Hitung leukosit ($10^3$/mL) | >40 | | 20-39.9 | 15-19.9 | 3-14.9 | | 1-2.9 | | <1 |

**Skor Umur**

| Umur | Skor |
|---|---|
| <44 | 0 |
| 45-54 | 2 |
| 55-64 | 3 |
| 65-74 | 5 |
| >75 | 6 |

**Skor Kondisi Kesehatan Kronis**

| Riwayat Insufisiensi Organ Berat | Skor |
|---|---|
| Pasien non-bedah | 5 |
| Pasien pascabedah darurat | 5 |
| Pasien pascabedah elektif | 2 |

Skor dapat dihitung di http://www.sfar.org/scores2/apache22.htm#calcul

**Tabel 2.** CT Scan Severity Index[25,31]

| CT Grade | Deskripsi | Poin | Nekrosis % | Nekrosis Poin | *Severity Index** |
|---|---|---|---|---|---|
| A | Pankreas normal | 0 | 0 | 0 | 0 |
| B | pankreas edematosa | 1 | 0 | 0 | 1 |
| C | B plus perubahan ekstrapankreatik ringan | 2 | <30 | 2 | 4 |
| D | perubahan ekstrapankreatik berat plus satu timbunan cairan | 3 | 30-50 | 4 | 7 |
| E | timbunan cairan multipel atau berlebihan | 4 | >50 | 6 | 10 |

**Skoring: poin *CT Grade* ditambah poin persentase nekrosis**

**Tabel 3.** Sistem skoring Imrie[26]

| Usia | > 55 tahun |
|---|---|
| Hitung leukosit | > 15.000 per $mm^3$ |
| Glukosa darah | > 180 mg/dL pada non-penderita diabetes |
| Laktat dehidrogenase serum | > 600 U/L |
| AST atau ALT serum | > 100 U/L |
| Kalsium serum | < 8 mg/dL |
| $PaO_2$ | < 60 mmHg |
| Albumin serum | < 3,2 g/dL |
| Urea serum | > 45 mg/dL |

**Skoring: 1 poin untuk setiap kriteria yang terpenuhi pada 48 jam setelah masuk**

**Tabel 4.** Kriteria Ranson[27]

| Pada saat masuk: | |
|---|---|
| ▪ Usia | > 55 tahun |
| ▪ Hitung leukosit | > 16.000 per $mm^3$ |
| ▪ Glukosa darah | > 200 mg/dL pada non-penderita diabetes |
| ▪ LDH serum | > 350 U/L |
| ▪ AST | > 250 U/L |
| **Dalam 48 jam pertama:** | |
| ▪ Penurunan hematokrit | > 10% |
| ▪ Peningkatan nitrogen urea darah | > 5 mg/dL |
| ▪ Kalsium serum | < 8 mg/dL |
| ▪ Defisit basa | > 4 mmol/L (4 mEq/L) |
| ▪ Sekuestrasi cairan | > 6000 mL |
| ▪ $PaO_2$ | < 60 mmHg |

**Skoring: 1 poin untuk setiap kriteria yang terpenuhi**

BUN: *blood urea nitrogen*

**Tabel 5.** *Glasgow scoring system* untuk prediksi keparahan pankreatitis akut.[28]

| No. | Parameter | Nilai *cut-off* |
|---|---|---|
| 1. | $PaO_2$ arterial | <60 mmHg |
| 2. | Albumin serum | <32 g/L |
| 3. | Kalsium serum | <2.0 mmol/L |
| 4. | Hitung leukosit | >15 x $10^9$/L |
| 5. | AST | >200 μ/L |
| 6. | LDH | >600 IU/L |
| 7. | Glukosa darah | >10 mmol/L (non-diabetik) |
| 8. | Urea plasma | >16 mmol/L |

**Skor: Lebih dari 3 kriteria positif merupakan indikasi pankreatitis akut berat.**

AST: aspartat aminotransferase; LDH: laktat dehidrogenase

**Tabel 6.** Sistem skor BISAP[29]

| No | Parameter |
|---|---|
| 1. | BUN > 25 mg/dL |
| 2. | Gangguan status mentalis (Skor GCS <15) |
| 3. | SIRS, yaitu 2 dari 4 hal berikut:<br>(1) Suhu <36°C atau >38°C<br>(2) Frekuensi napas >20/menit atau $P_aCO_2$ <32 mmHg<br>(3) Denyut nadi >90 x/ menit<br>(4) Leukosit <4.000 atau >12.000 sel/mm3 atau >10% sel imatur |
| 4. | Usia >60 tahun |
| 5. | Efusi pleura terdeteksi dari pencitraan |

**Skor: satu poin untuk setiap variabel dalam 24 jam pertama.**

BISAP: *bedside index for severity in acute pancreatitis;* SIRS: *systemic inflammatory response syndrome.*

**Tabel 7.** Komponen HAPS[30]

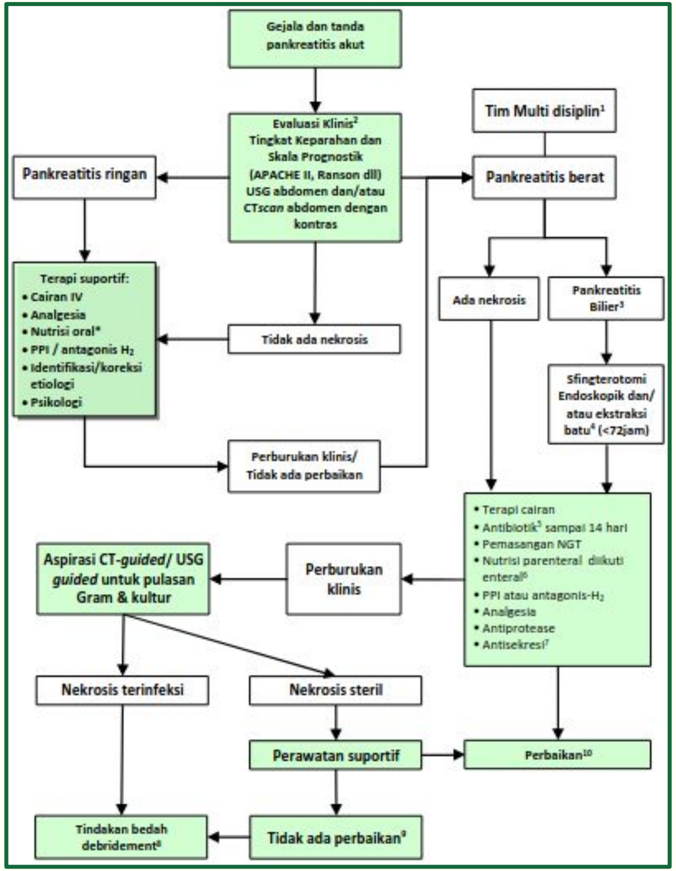
| No | Parameter |
|---|---|
| 1 | Nyeri lepas *(rebound tenderness)* atau perut papan *(guarding)*: ada/ tidak |
| 2 | Kadar hematokrit: normal atau meningkat |
| 3 | Kadar kreatinin serum: normal atau meningkat |

**Interpretasi: kondisi pasien dianggap tidak berbahaya atau *"harmless"* jika tidak ada *rebound tenderness/ guarding*, hematokrit normal dan kreatinin serum normal.**

HAPS: *harmless acute pancreatitis score*

Manan HC, Makmun HD, Aulia C. Konsensus nasional penatalaksanaan pankreatitis akut di Indonesia. Perkumpulan Gastroenterologi Indonesia (PGI). 2011. https://pbpgigastro.com/wp-content/uploads/2024/07/Konsensus-Nasional-Penatalaksanaa-Pankreatitis-Akut-di-Indonesia-Tahun-2011.pdf

# Manajemen Pankreatitis Akut




Manan HC, Makmun HD, Aulia C. Konsensus nasional penatalaksanaan pankreatitis akut di Indonesia. Perkumpulan Gastroenterologi Indonesia (PGI). 2011. https://pbpgigastro.com/wp-content/uploads/2024/07/Konsensus-Nasional-Penatalaksanaa-Pankreatitis-Akut-di-Indonesia-Tahun-2011.pdf

# Terapi Suportif Umum

| |
|---|
| Resusitasi Cairan:<br>Cairan intravena (kristaloid / koloid) → mengatasi hipovolemia akibat hilangnya cairan (muntah dan diaforesis)<br>Pemantauan tanda vital, pengeluaran urin (minimal 0,5 mL/kgBB/jam), penurunan hematokrit selama 12–24 jam |
| Oksigen:<br>Untuk menjaga saturasi arterial ≥95%, terutama 24–48 jam pertama → mengurangi risiko gagal organ |
| Koreksi Elektrolit dan Metabolik:<br>Penyesuaian elektrolit serta perbaikan gangguan metabolik dilakukan secepatnya |
| Manajemen Nyeri:<br>Meperidin, morfin atau hydromorphone dapat diberikan (perhatian potensi efek samping → akumulasi metabolit normeperidin dari meperidin) |
| Nutrisi:<br>• Pemberian nutrisi enteral dalam 48 jam pertama (melalui nasojejunal tube) dianjurkan pada pasien dengan malnutrisi / pemberian makan melalui mulut tidak memungkinkan |

Manan HC, Makmun HD, Aulia C. Konsensus nasional penatalaksanaan pankreatitis akut di Indonesia. Perkumpulan Gastroenterologi Indonesia (PGI). 2011. https://pbpgigastro.com/wp-content/uploads/2024/07/Konsensus-Nasional-Penatalaksanaa-Pankreatitis-Akut-di-Indonesia-Tahun-2011.pdf

# Terapi untuk Membatasi Komplikasi

| |
|---|
| Mengistirahatkan Pankreas:<br>Dilakukan dengan cara puasa (NPO) → mengurangi stimulasi sekresi pankreas<br>Obat-obatan: antagonis reseptor-H2, proton pump inhibitor (pantoprazol), atropin, agen anti-sekresi seperti somatostatin dan octreotide |
| Mengurangi Aktivitas Protease:<br>Beberapa pendekatan melibatkan pemberian antiprotease (aprotinin), fresh frozen plasma atau bilasan peritoneal → mengurangi aktivitas enzim proteolitik |
| Mencegah SIRS:<br>Upaya untuk mencegah respon inflamasi sistemik (SIRS) di antaranya melalui penggunaan antagonis terhadap mediator inflamasi seperti platelet-activating factor (PAF), meskipun intervensi dengan lexipafant belum menunjukkan manfaat klinis yang signifikan |
| Pengangkatan Batu di Common Bile Duct (CBD):<br>Pankreatitis bilier → pengangkatan batu menggunakan ERCP dan sfingterotomi dianjurkan, terutama bila terdapat risiko tinggi komplikasi (gagal organ / kolangitis)<br>Idealnya ERCP dilakukan dalam 72 jam pertama setelah timbulnya nyeri |

Manan HC, Makmun HD, Aulia C. Konsensus nasional penatalaksanaan pankreatitis akut di Indonesia. Perkumpulan Gastroenterologi Indonesia (PGI). 2011. https://pbpgigastro.com/wp-content/uploads/2024/07/Konsensus-Nasional-Penatalaksanaa-Pankreatitis-Akut-di-Indonesia-Tahun-2011.pdf

3

# Somatostatin

# Somatostatin

Ditemukan 1972 oleh Roger Guillemin (pemenang Nobel)

Hormon somatostatin alami diproduksi otak & pankreas

Fungsi:
Inhibisi hormon insulin, growth hormone (GH), asam lambung,
Menyempitkan pembuluh pencernaan / splanik→ menurunkan hipertensi portal

Baranwal A, Rastogi P, Dixit A. Roger Guillemin: A Century of Life, A Legacy of Endocrine Discoveries. Cureus. 2024;16(9):e68540. Published 2024 Sep 3. doi:10.7759/cureus.68540
Pless J. The history of somatostatin analogs. J Endocrinol Invest. 2005;28(11 Suppl International):1-4.

Theodoropoulou M, Stalla GK. Somatostatin receptors: From signaling to clinical practice. Frontiers in Neuroendocrinology.2023;34(3):228–252. https://doi.org/10.1016/j.yfrne.2013.07.005

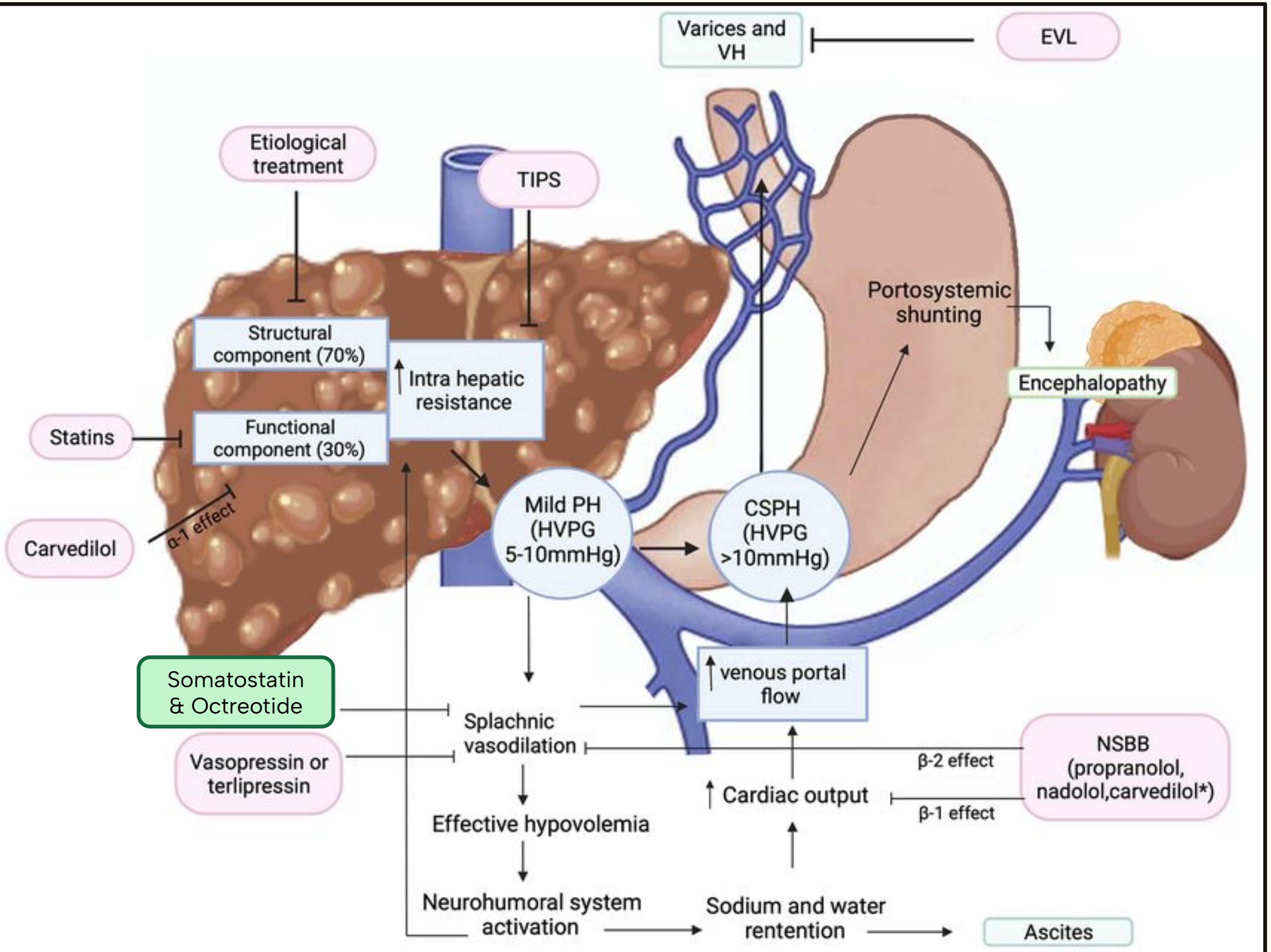
# Somatostatin dan Afinitas Reseptor Somatostatin (SSTR)

| | |
|---|---|
| 2 bentuk biologis somatostatin:<br>Somatostatin-14 & -28, | Analogue Somatostatin:<br>Octreotide<br>Lancreotide |
| Keduanya berikatan dengan reseptor somatostatin (SSTR) 1–4 dengan afinitas lebih tinggi | |

Affinity of human, mouse, and rat SRIF-14 and SRIF-28, lanreotide, octreotide and SOM230 for human (h), rat (r) and/or mouse (m) SSTR1, 2, 3, 4, and 5. The species (h, r, m) and unit (pKi, pIC50, pEC50) are indicated in brackets. Data obtained from Bruns et al. (1996), Hoyer et al. (2000), and Schmid and Schoeffter (2004).

| | SSTR1 | SSTR2 | SSTR3 | SSTR4 | SSTR5 |
|---|---|---|---|---|---|
| SRIF-14 | 8.6–9.5 (h, pKi)<br>8.4 (r, pIC50) | 8.9–10.5 (h, pKi)<br>8.8 (r, pKi)<br>9.14–8.99 (r, pEC50)<br>8.8–9.6 (m, pKi) | 8.7–10.0 (h, pKi)<br>8.9 (r, pKi)<br>10.1 (m, pIC50) | 8.4–9.3 (h,pKi)<br>8.2 (r, pKi) | 8.4–9.9 (h, pKi)<br>8.4 (r, pKi)<br>7.7–9.1 (m, pKi) |
| SRIF-28 | 8.6–9.4 (h, pKi) | 8.4–10.2 (h, pKi)<br>8.8 (r, pKi)<br>9.0–9.5 (m, pKi) | 8.1–9.9 (h, pKi)<br>9.1 (r, pKi)<br>10.2 (m, pKi) | 8.1–9.2(h, pKi) | 9.2–10.3(h, pKi)<br>9.0 (r, pKi)<br>9.8–9.9(m, pKi) |
| Lanreotide | – | 8.7–9.6(h, pKi)<br>8.8 (r, pKi)<br>6.9–9.1 (m, pKi) | 7.2–8.0 (h, pKi)<br>6.9 (r, pKi)<br>8.2 (m, pIC50) | – | 7.4–9.3 (h, pKi)<br>7.6 (m, pKi) |
| Octreotide | – | 8.7–9.9 (h, pKi)<br>8.7 (r, pKi)<br>7.5 – 9.2 (m, pKi) | 7.4–8.6, h, pKi)<br>7.8 (r, pKi)<br>8.5 (m, pIC50) | – | 7.2–9.9 (h, pKi)<br>7.8 (r, pKi)<br>7.7–9.9 (m, pKi) |
| SOM-230 | 8.0(h, pIC50) | 9.0 (h, pIC50)<br>7.76 – 9.07 (r, pEC50) | 8.8 (h, pIC50) | – | 9.8 (h, pIC50) |

Diaz-Soto MP, et al. Management of varices and variceal hemorrhage in liver cirrhosis: a recent update. Ther Adv Gastroenterol.2022;15:1–12. http://dx.doi.org/10.1177/17562848221101712




# Mechanism of Action

Somatostatin

↓

Mengikat reseptor SST2 arteri splanknik

↓

Vasokonstriksi Splanknik

↓

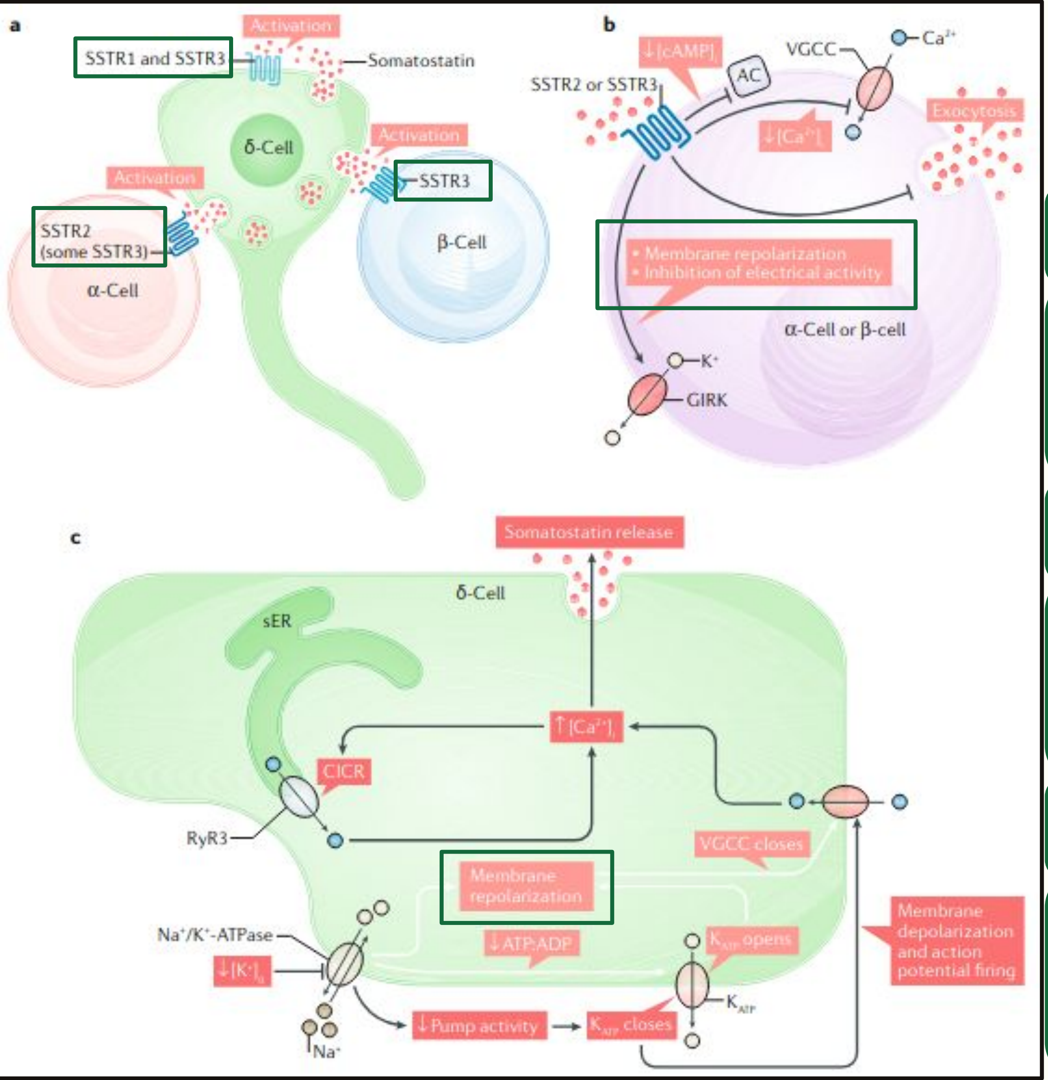
Menurunkan aliran portal & HVPG

↓

Penurunan venous portal flow

↓

Menekan risiko varises pecah & bleeding




Rorsman, P., Huising, M.O. The somatostatin-secreting pancreatic δ-cell in health and disease. Nat Rev Endocrinol 14, 404–414 (2018). https://doi.org/10.1038/s41574-018-0020-6

# Mechanism of Action

## a. Reseptor Somatostatin di Pulau Langerhans

Sel δ melepaskan somatostatin ke ruang antar-sel
Autokrin: somatostatin aktifkan SSTR1/3 pada sel δ
Parakrin: aktifkan SSTR2 pada sel α (sebagian SSTR3)
Parakrin: aktifkan SSTR3 pada sel β
**Aktivasi reseptor menekan sekresi glukagon & insulin**

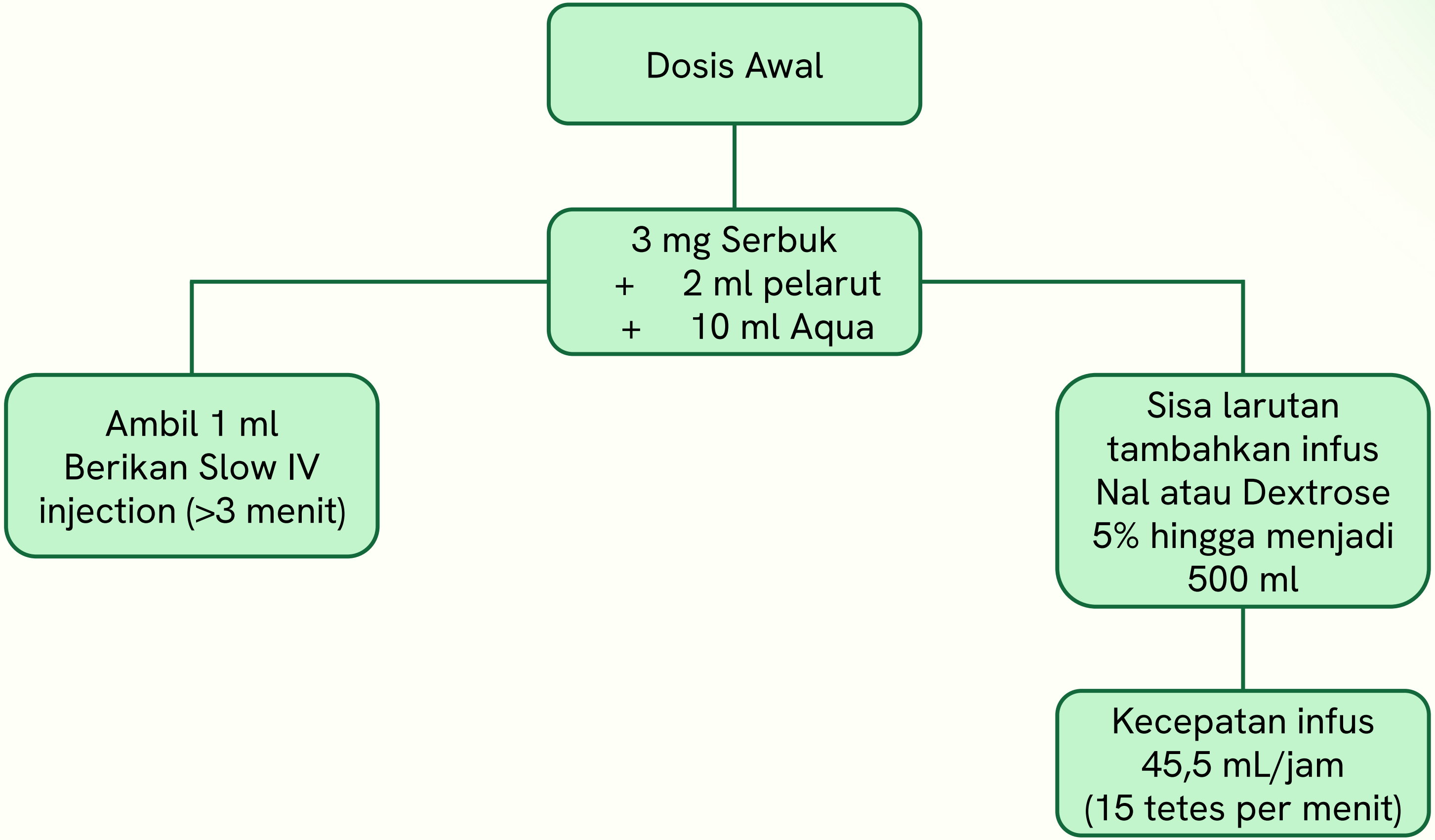
## b. Sinyal Downstream & Penghambatan Hormon

SSTR2/3 → Gi → adenilat siklase terhambat
cAMP turun → eksositosis hormon α/β berkurang
VGCC tertutup → $Ca^{2+}$ sitosolik menurun
Kanal GIRK terbuka → repolarisasi membran cepat
**Inaktivasi listrik menghentikan eksositosis hormon**

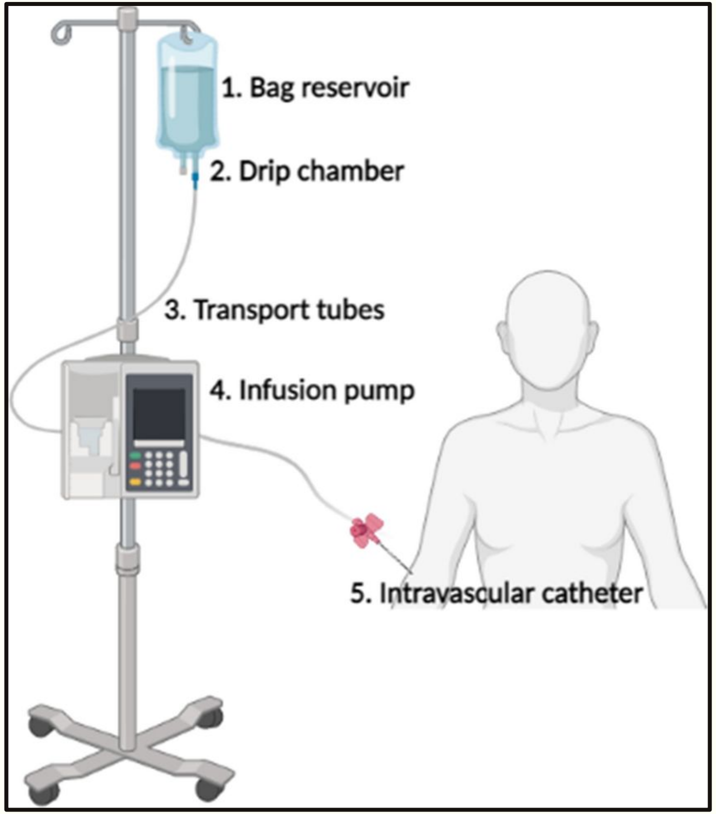
## c. Regulasi Sekresi Somatostatin oleh $K^+$

Hipokalemia menghambat $Na^+/K^+$-ATPase sel δ
Rasio ATP:ADP naik → KATP menutup
Depolarisasi membran membuka VGCC
$Ca^{2+}$ masuk; CICR RyR3 menaikkan $Ca^{2+}$ lanjut
$Ca^{2+}$ tinggi memacu pelepasan somatostatin baru
Somatostatin menginduksi repolarisasi

# Dosis Somatostatin



# Infusion Pump




Terapi perdarahan berat dengan Dosis Awal
3 mg Serbuk dilarutkan 2 ml pelarut
+10 ml Aqua = 12ml
Ambil 1 ml (250mcg), berikan Slow IV injection (>3 menit)
Sisa larutan tambahkan infus NaCl atau Dextrose 5%
hingga menjadi 500 ml
500ml = 2750mcg somatostatin
45,5ml = 250mcg

Tpm = (faktor tetes x volume) / (berapa jam x 60)
= [20 (macrodrip) x 45,5ml] / (1jam x 60)
= 15 tetes per menit

# Efektifitas & Keamanan Somatostatin + EVL / EST pada BEV

## A randomized controlled trial comparing ligation and sclerotherapy as emergency endoscopic treatment added to somatostatin in acute variceal bleeding ☆,☆☆

Càndid Villanueva[1,*], Marta Piqueras[1], Carles Aracil[1], Cristina Gómez[1], Josep M. López-Balaguer[1], Begoña Gonzalez[1], Adolfo Gallego[1], Xavier Torras[1], Germà Soriano[1], Sergio Sáinz[1], Salvador Benito[2], Joaquim Balanzó[1]

[1]*Gastrointestinal Bleeding Unit, Department of Gastroenterology, Hospital de la Santa Creu i Sant Pau, Barcelona, Spain*
[2]*Intermediate Care Unit, Hospital de la Santa Creu i Sant Pau, Barcelona, Spain*

***Background/Aims*:** **The currently recommended treatment for acute variceal bleeding is the association of vasoactive drugs and endoscopic therapy. However, which emergency endoscopic treatment combines better with drugs has not been clarified. This study compares the efficacy and safety of variceal ligation and sclerotherapy as emergency endoscopic treatment added to somatostatin.**

***Methods*:** **Patients admitted with acute gastrointestinal bleeding and with suspected cirrhosis received somatostatin infusion (for 5 days). Endoscopy was performed within 6 h and those with esophageal variceal bleeding were randomized to receive either sclerotherapy ($N = 89$) or ligation ($N = 90$).**

***Results*:** **Therapeutic failure occurred in 21 patients treated with sclerotherapy (24%) and in nine treated with ligation (10%) (RR = 2.4, 95% CI = 1.1–4.9). Failure to control bleeding occurred in 15% vs 4%, respectively ($P = 0.02$). Treatment group, shock and HVPG >16 mmHg were independent predictors of failure. Side-effects occurred in 28% of patients receiving sclerotherapy vs 14% with ligation (RR = 1.9, 95% CI = 1.1–3.5), being serious in 13% vs 4% ($P = 0.04$). Six-week survival probability without therapeutic failure was better with ligation ($P = 0.01$).**

***Conclusions*:** **The use of variceal ligation instead of sclerotherapy as emergency endoscopic therapy added to somatostatin for the treatment of acute variceal bleeding significantly improves the efficacy and safety.**

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**Outcomes in the two groups**

| | SMT + EST group ($N = 89$) | SMT + EVL group ($N = 90$) | RR (95% CI) | *P* value |
|---|---|---|---|---|
| Therapeutic failure[a] | | | | |
| Overall | 21 (24%) | 9 (10%) | 2.4 (1.1–4.9) | 0.02 |
| Child–Pugh class A/B | 14/68 (21%) | 3/66 (4%) | 4.5 (1.4–15.0) | |
| Child–Pugh class C | 7/21 (33%) | 6/24 (25%) | 1.3 (0.5–3.3) | |
| Active bleeding[b] | 5/21 (24%) | 3/17 (18%) | 1.3 (0.4–4.8) | |
| No active bleeding | 16/68 (23%) | 6/73 (8%) | 2.9 (1.2–6.9) | |
| Hypovolemic shock[c] | 12/26 (46%) | 6/32 (19%) | 2.5 (1.1–5.7) | |
| Absence of hypovolemic shock | 9/63 (14%) | 3/58 (5%) | 2.8 (0.9–9.7) | |
| Failure to control acute bleeding episode | | | | |
| Overall | 13 (15%) | 4 (4%) | 3.3 (1.1–9.7) | 0.02 |
| Active bleeding[b] | 5/21 (24%) | 2/17 (12%) | 2.2 (0.5–9.2) | |
| Hypovolemic shock[c] | 9/26 (35%) | 3/32 (9%) | 3.7 (1.1–12.2) | |
| Time from admission to cessation of bleeding (h)[d,e] | $8.4 \pm 6.2$ | $6.5 \pm 5.8$ | | 0.05 |
| Early rebleeding[f] | 7/76 (9%) | 4/86 (5%) | 1.2 (0.6–6.5) | 0.25 |
| Late rebleeding (5–42 days) | 11 (12%) | 6 (7%) | 1.8 (0.7–4.8) | 0.21 |
| Transfusion during the trial period (units of packed red cells) | | | | 0.05 |
| Mean[d] | $3.9 \pm 3.0$ | $3.1 \pm 2.3$ | | |
| Median | 3 | 3 | | |
| Range | 0–12 | 0–11 | | |
| Days in hospital[d] | $15 \pm 9$ | $13 \pm 7$ | | 0.27 |
| Mortality[a] | | | | |
| Death within 5 days | 3 (3%) | 3 (3%) | 1.01 (0.2–4.9) | 0.72 |
| Death within 42 days | 19 (21%) | 12 (13%) | 1.6 (0.8–3.1) | 0.17 |

[a] Death within 5 days with the hemorrhage under control occurred in 1 case in each group.
[b] Presence of active bleeding during endoscopy.
[c] Presence of hypovolemic shock at admission. Hypovolemic shock was defined as systolic blood pressure <100 mmHg and pulse rate >100 bpm.
[d] Plus-minus values are means ± SD.
[e] Time from admission to cessation of bleeding was defined as the hours from admission to the first hospital (time 0) to the first clear gastric lavage, which occurred in patients with initial control of bleeding.
[f] Early rebleeding was due to esophageal varices in all patients who had repeated endoscopy. Only in 1 patient (of the SMT + EST group) endoscopy was not performed.

**Somatostatin + EVL / Sclerotherapy meningkatkan efikasi dan keamanan secara signifikan**

Càndid Villanueva; Marta Piqueras; Carles Aracil; Cristina Gómez; Josep M. López-Balaguer; Begoña Gonzalez; Adolfo Gallego; Xavier Torras; Germà Soriano; Sergio Sáinz; Salvador Benito; Joaquim Balanzó. (2006). A randomized controlled trial comparing ligation and sclerotherapy as emergency endoscopic treatment added to somatostatin in acute variceal bleeding. , 45(4), 0–567. doi:10.1016/j.jhep.2006.05.016

# Multicenter randomized controlled trial comparing different schedules of somatostatin in the treatment of acute variceal bleeding

Eduardo Moitinho[1], Ramon Planas[2], Rafael Bañares[3], Agustín Albillos[4], Luis Ruiz-del-Arbol[5], Carmen Gálvez[6], Jaime Bosch[1,*], Variceal Bleeding Study Group[1]

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[2]*Department of Gastroenterology, Hospital Germans Trias i Pujol, University Autónoma of Barcelona, Badalona, Spain*
[3]*Department of Gastroenterology, Hospital General Gregorio Marañón, Universidad Complutense, Madrid, Spain*
[4]*Department of Gastroenterology, Clínica Puerta de Hierro, University of Alcalá de Henares, Madrid, Spain*
[5]*Department of Gastroenterology, Hospital Ramón y Cajal, University of Alcalá de Henares, Madrid, Spain*
[6]*Department of Gastroenterology, Hospital Reina Sofía, University of Córdoba, Córdoba, Spain*

Eduardo Moitinho; Ramon Planas; Rafael Bañares; Agustín Albillos; Luis Ruiz-del-Arbol; Carmen Gálvez; Jaime Bosch. (2001). Multicenter randomized controlled trial comparing different schedules of somatostatin in the treatment of acute variceal bleeding. , 35(6), 712–718. doi:10.1016/s0168-8278(01)00206-9

# Efektifitas Somatostatin pada BEV

***Background/Aims*: The dose of somatostatin used for variceal bleeding (250 μg/h) is lower than that proven to effectively decrease portal pressure and azygos blood flow (500 μg/h). Moreover, i.v. somatostatin boluses have greater effects than continuous infusions. The aim of this study was to investigate whether higher doses of somatostatin and repeated boluses may increase its efficacy in controlling variceal bleeding.**

***Methods*: A total of 174 patients with acute variceal bleeding were randomized to receive for 48 h: (A) one 250 μg bolus + 250 μg/h infusion; (B) three 250 μg boluses + 250 μg/h infusion; (C) three 250 μg boluses + 500 μg/h infusion.**

***Results*: The three schedules of somatostatin were equally effective in controlling variceal bleeding (73, 75 and 81%, respectively, NS). Multivariate analysis showed active bleeding at endoscopy ($n = 75$) as the only predictor of failure to control bleeding. In these patients, the 500 μg/h infusion dose achieved a higher rate of control of bleeding (82 vs. 60%, $P < 0.05$), less transfusions (3.7 ± 2.7 vs. 2.5 ± 2.3 UU, $P = 0.07$) and better survival (93 vs. 70%, $P < 0.05$) than schedules A/B.**

***Conclusions*: Somatostatin is highly effective in controlling variceal bleeding. Patients with active bleeding at emergency endoscopy may benefit from higher doses of somatostatin infusion.**

Control of bleeding, early re-bleeding and mortality: results of treatment in high risk patients, with active bleeding at emergency endoscopy

| Patients with active bleeding at endoscopy | Schedule A ($n = 24$) (95% CI) | Schedule B ($n = 23$) (95% CI) | Schedule C ($n = 28$) (95% CI) | $P$ value |
|---|---|---|---|---|
| Control of bleeding (%) | 58 | 61 | 82 | 0.13 |
| Transfusions (UU) | 4.0 ± 2.4 (2.9–5.0) | 3.4 ± 3.0 (2.0–4.7) | 2.5 ± 2.3 (1.6–3.4) | 0.14 |
| Early re-bleeding (%) | 25 | 15 | 15 | NS |
| 6-week mortality (%) | 33 | 26 | 7 | 0.05 |

RCT multisenter 174 pasien:

- A 250 μg/h
- B + 3 bolus 250 μg
- C 500 μg/h 48 jam

**Hemostasis**: 73 %, 75 %, 81 % (non-signifikan)
**Bleed aktif**: 500 μg/h sukses 82 % vs 60 %
**Mortalitas**: 7 % vs 30 %

**Somatostatin efektif menghentikan perdarahan dan lebih cepat pada dosis tinggi, dengan efek samping ringan**

AP&T Alimentary Pharmacology and Therapeutics

# Meta-analysis: vasoactive medications for the management of acute variceal bleeds

M. Wells, N. Chande, P. Adams, M. Beaton, M. Levstik, E. Boyce & M. Mrkobrada

**Table 2** | Summary of findings for the comparative meta-analysis of vasoactive medications

| Outcome | Number of trials | Number of patients | Combined RR (95% CI) | *P*-value |
|---|---|---|---|---|
| Terlipressin compared with somatostatin | | | | |
| Mortality | 7 | 635 | 1.05 (0.74–1.49) | 0.78 |
| Haemostasis | 7 | 635 | 1.01 (0.94–1.07) | 0.84 |
| Rebleeding | 5 | 542 | 1.07 (0.74–1.54) | 0.73 |
| Terlipressin compared with vasopressin | | | | |
| Mortality | 5 | 290 | 1.36 (0.94–1.96) | 0.10 |
| Haemostasis | 5 | 247 | 1.08 (0.85–1.39) | 0.53 |
| Rebleeding | 5 | 262 | 1.41 (0.92–2.18) | 0.12 |
| Octreotide compared with terlipressin | | | | |
| Mortality | 6 | 752 | 0.98 (0.63–1.52) | 0.91 |
| Haemostasis | 7 | 782 | 1.00 (0.91–1.10) | 0.98 |
| Rebleeding | 5 | 375 | 1.10 (0.73–1.68) | 0.65 |
| Hospital stay | 2 | 412 | 1.25 (0.54–3.04) | 0.17 |
| Transfusion requirement | 4 | 527 | −0.49 (−1.52 to 0.55) | 0.36 |
| Octreotide compared with somatostatin | | | | |
| Mortality | 2 | 173 | 1.51 (0.60–3.79) | 0.38 |
| Haemostasis | 3 | 314 | 0.89 (0.76–1.05) | 0.17 |
| Rebleeding | 2 | 173 | 1.66 (0.91–3.03) | 0.10 |
| Octreotide compared with vasopressin | | | | |
| Mortality | 1 | 48 | 1.00 (0.54–1.85) | 1.00 |
| Haemostasis | 2 | 204 | 1.31 (1.10–1.56) | 0.002 (favours octreotide) |
| Vasopressin compared with somatostatin | | | | |
| Mortality | 4 | 207 | 1.25 (0.69–2.28) | 0.47 |
| Haemostasis | 6 | 379 | 0.75 (0.61–0.92) | 0.006 (favours somatostatin) |
| Rebleeding | 3 | 158 | 0.60 (0.31–1.14) | 0.12 |

# Efektifitas Somatostatin pada BEV

**Somatostatin vs Agen Vasoaktif**

Somatostatin vs Octreotide:

- Hemostasis RR 0,89 (non-signifikan)
- Rebleed RR 1,66 (non-signifikan)

Somatostatin vs Terlipressin:

- Hemostasis RR 1,01 (non-signifikan)
- Mortalitas RR 1,05 (non-signifikan)
- Rebleeding 5 RR 1,07 (non-signifikan)

Somatostatin vs Vasopressin:

- Hemostasis lebih baik RR 0,75 pada somatostatin

Wells M, Chande N, Adams P, et al. Meta-analysis: vasoactive medications for the management of acute variceal bleeds. Aliment Pharmacol Ther. 2012;35(11):1267-1278. doi:10.1111/j.1365-2036.2012.05088.x

# Comparison of drugs facilitating endoscopy for patients with acute variceal bleeding: a systematic review and network meta-analysis

Ziyuan Zou[1,2#], Xinwen Yan[1,2#], Huanpeng Lu[2#], Xingshun Qi[3], Ye Gu[4], Xun Li[1], Bin Wu[5], Xiaolong Qi[1]

**Table 3** SUCRA value, probability and mean rank for each treatment

| Outcomes | SUCRA value (%) | Probability best (%) | Mean rank |
|---|---|---|---|
| Six-week mortality | | | |
| Somatostatin | 53.1 | 24.8 | 2.4 |
| Octreotide | 61.9 | 24.8 | 2.1 |
| Vasopressin | 52.7 | 42.2 | 2.4 |
| Terlipressin | 32.2 | 8.2 | 3.0 |
| Five-day mortality | | | |
| Somatostatin | 71.0 | 30.6 | 2.2 |
| Octreotide | 60.9 | 17.3 | 2.6 |
| Vasopressin | 35.4 | 26.9 | 3.6 |
| Terlipressin | 62.9 | 24.1 | 2.5 |
| Placebo | 19.8 | 1.1 | 4.2 |
| Five-day rebleeding | | | |
| Somatostatin | 49.4 | 8.1 | 3.0 |
| Octreotide | 57.4 | 16.2 | 2.7 |
| Vasopressin | 61.2 | 37.4 | 2.6 |
| Terlipressin | 71.9 | 38.2 | 2.1 |
| Placebo | 10.1 | 0.1 | 4.6 |
| Control of initial bleeding | | | |
| Somatostatin | 60.7 | 23.8 | 2.6 |
| Octreotide | 80.5 | 45.3 | 1.8 |
| Vasopressin | 58.5 | 23.8 | 2.7 |
| Terlipressin | 49.7 | 7.1 | 3.0 |
| Placebo | 0.6 | 0.0 | 5.0 |

**Table 2** Summary of findings reporting the comparative efficacy of pharmacological agents

| Outcomes | Studies, n | Pairwise meta-analysis OR (95% CI) | Network meta-analysis OR (95% CI) |
|---|---|---|---|
| Somatostatin compared with octreotide | | | |
| Six-week mortality | 2 | 1.04 (0.61–1.75) | 1.05 (0.53–2.24) |
| Five-day mortality | 1 | 1.00 (0.55–1.84) | 0.94 (0.45–1.93) |
| Five-day rebleeding | 1 | 1.11 (0.46–2.66) | 1.12 (0.21–5.83) |
| Control of initial bleeding | 1 | 0.96 (0.57–1.63) | 0.88 (0.29–3.49) |
| Somatostatin compared with vasopressin | | | |
| Six-week mortality | NA | NA | 1.06 (0.25–4.79) |
| Five-day mortality | 1 | 0.55 (0.05–6.59) | 0.53 (0.02–7.96) |
| Five-day rebleeding | 1 | 1.20 (0.29–4.95) | 1.13 (0.15–8.78) |
| Control of initial bleeding | 1 | 2.85 (0.27–29.84) | 1.14 (0.34–5.95) |
| Somatostatin compared with terlipressin | | | |
| Six-week mortality | 1 | 0.87 (0.52–1.48) | 0.87 (0.42–1.87) |
| Five-day mortality | 1 | 1.11 (0.60–2.07) | 0.95 (0.39–2.00) |
| Five-day rebleeding | 1 | 1.40 (0.55–3.55) | 1.47 (0.27–8.47) |
| Control of initial bleeding | 1 | 0.82 (0.48–1.41) | 0.88 (0.22–2.69) |
| Somatostatin compared with placebo | | | |
| Six-week mortality | NA | NA | NA |
| Five-day mortality | 2 | 0.51 (0.24–1.08) | 0.63 (0.30–1.33) |
| Five-day rebleeding | NA | NA | 0.42 (0.06–2.64) |
| Control of initial bleeding | NA | NA | 5.15 (1.40–27.39) |

**Table 4** Adverse events

| Interventions | Total adverse events, n (%) | Serious adverse events, n (%) | Total patients, (n) |
|---|---|---|---|
| Somatostatin | 59 (14.4) | 2 (0.5) | 408 |
| Octreotide | 30 (9.1) | 0 (0.0) | 330 |
| Vasopressin | 60 (57.1) | 7 (6.7) | 105 |
| Terlipressin | 106 (24.6) | 3 (0.7) | 431 |

# Efektifitas & Keamanan Somatostatin pada BEV

**Efektivitas Somatostatin pada Acute Variceal Bleeding**:

- Hemostasis awal meningkat (OR 5,15 vs plasebo)
- Peringkat 2 efektivitas hemostasis (SUCRA 60,7 %)
- Terbaik pada mortalitas 5 hari (SUCRA 71 %)
- Efek samping total 14,4 %; SAE 0,5 % — lebih aman dari vasopresin/terlipressin

Zou Z, Yan X, Lu H, et al. Comparison of drugs facilitating endoscopy for patients with acute variceal bleeding: a systematic review and network meta-analysis. Ann Transl Med. 2019;7(23):717. doi:10.21037/atm.2019.12.26

*Gut*, 1989, **30**, 223–227

# Somatostatin in the treatment of acute pancreatitis: a prospective randomised controlled trial

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*From the Department of Surgery, University of Hong Kong, Queen Mary Hospital, Hong Kong*

Table 4 *Laboratory tests*

| | | *Control group* | *Somatostatin group* | |
|---|---|---|---|---|
| n | | 36 | 35 | p |
| Age | yr | 62·57 (13·20) | 59·74 (21·86) | NS |
| WBC | $10^9$/l | 14·67 (7·22) | 13·97 (6·73) | NS |
| WBC 2* | | 12·71 (6·45) | 9·45 (4·06) | 0·0329 |
| Amylase | IU/l | 875 (395) | 1069 (487) | NS |
| Amylase 2 | | 170 (216) | 138 (123) | NS |
| SGOT | IU/l | 206 (251) | 219 (371) | NS |
| SGOT 2 | | 51 (48) | 61 (78) | NS |
| LDH | IU/l | 361 (265) | 386 (345) | NS |
| LDH 2 | | 311 (150) | 216 (107) | 0·0088 |
| Urea | mmol/l | 6·17 (2·23) | 6·33 (2·71) | NS |
| Urea 2 | | 5·64 (4·24) | 4·27(3·27) | 0·0338 |
| Glucose | mmol/l | 8·70 (6·02) | 7·62 (2·71) | NS |
| Glucose 2 | | 6·88 (2·59) | 7·65 (2·52) | NS |
| Calcium | mmol/l | 2·16 (0·21) | 2·14 (0·20) | NS |
| Calcium 2 | | 2·02 (0·18) | 1·97 (0·16) | NS |
| Albumen | mmol/l | 43·30 (5·23) | 42·40 (6·93) | NS |
| Albumen 2 | | 36·75 (5·48) | 36·29 (5·58) | NS |
| $PaO_2$ | mmHg | 61·97 (17·46) | 68·22 (17·80) | NS |
| $PaO_2$ 2 | | 63·78 (14·98) | 68·90 (21·16) | NS |

*WBC 2 WBC at two days after admission.
( ) standard deviation.

Table 5 *Complications*

| *Complication* | *Control group (n)* | *Somatostatin group (n)* |
|---|---|---|
| Pseudocyst | 1 | 1 |
| Pancreatic abscess | 1 | 0 |
| Inflammatory pancreatic swelling | 6 | 2 |
| Upper gastrointestinal haemorrhage | 3 | 1 |
| Liver abscess | 1 | 0 |
| Multiorgan failure | 1 | 1 |
| | 13 | 5 |

Table 3 *Aetiology of acute pancreatitis*

| | |
|---|---|
| Gall stones | 37 |
| Alcohol | 11 |
| Ascaris | 1 |
| Unknown | 22 |

# Efektifitas & Keamanan Somatostatin pada Pankeratitis Akut

**Somatostatin IV (bolus 250 µg diikuti infus 100 µg/jam selama 48 jam) vs kontrol (plasebo)**

- **Penurunan signifikan kadar leukosit, LDH, dan ureum setelah 2 hari pada kelompok somatostatin dibanding kontrol**
- **Insidensi komplikasi lokal cenderung lebih rendah pada kelompok somatostatin, meskipun tidak signifikan secara statistik**
- **Mortalitas rendah dan serupa di kedua kelompok (masing-masing ~3% vs ~6%)**

Choi TK, Mok F, Zhan WH, Fan ST, Lai EC, Wong J. Somatostatin in the treatment of acute pancreatitis: a prospective randomised controlled trial. Gut. 1989;30(2):223-227. doi:10.1136/gut.30.2.223

# Influence of Somatostatin in the Evolution of Acute Pancreatitis

*A Prospective Randomized Study*

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*Hospital de Tarragona Joan XXIII. Facultad de Medicina, Department of Surgery, University of Barcelona*

Table 2
Clinical Evolution of the Groups

| | Control group | Somatostatin group |
|---|---|---|
| Length of hospital stay (d) | 20.28 | 14.92 |
| Pancreatic surgery (*p*) | 9 | 4 |
| Mortality (*p*) | 1 | 1 |

Table 3
Evolution CT Images According to Ranson Signs

| | Control group | | Somatostatin group | |
|---|---|---|---|---|
| Grade | CT1 | CT2 | CT1 | CT2 |
| A | 16 | 17 | 7 | 15 |
| B | 7 | 6 | 18 | 17 |
| C | 11 | 10 | 6 | 4 |
| D | 12 | 8 | 17 | 11 |
| E | 4 | 9 | 2 | 3 |
| Overall evolution | | | | |
| CT2 improvement | | 5 | | 19 |
| CT2 worsening | | 7 | | 4 |
| CT2 no changes | | 38 | | 27 |

Table 4
Number of Patients with CT Changes

| Control group | | Somatostatin group | |
|---|---|---|---|
| No. pat. | CT1/CT2 change | No. pat. | CT1/CT2 change |
| 1 | A......C | 1 | A......B |
| 1 | A......D | 9 | B......A |
| 2 | B......A | 3 | C......B |
| 1 | C......A | 1 | C......D |
| 1 | C......B | 4 | D......B |
| 1 | C......D | 2 | D......C |
| 1 | D......C | 2 | D......E |
| 4 | D......E | 1 | E......D |

# Efektifitas Somatostatin pada Pankreatitis Akut

**Somatostatin IV 250 µg/jam 48 jam (setelah bolus 250 µg) vs kontrol (tanpa somatostatin)**

- **Perbaikan signifikan pada temuan CT scan pankreas dalam 48 jam pada kelompok somatostatin dibanding kontrol**
- **Rata-rata lama rawat lebih pendek**

Luengo L, Vicente V, Gris F, et al. Influence of somatostatin in the evolution of acute pancreatitis. A prospective randomized study. Int J Pancreatol. 1994;15(2):139-144. doi:10.1007/BF02924664

# Efektifitas Somatostatin pada Pankreatitis Akut

Intensive Care Med (1998) 24: 37–39
© Springer-Verlag 1998

ORIGINAL

M. Planas
A. Pérez
R. Iglesia
I. Porta
J. R. Masclans
B. Bermejo

## Severe acute pancreatitis: treatment with somatostatin

**Table 2** Main characteristics of the patients

| | Control group | Somatostatin group | p |
|---|---|---|---|
| No. | 22 | 24 | |
| Age (years) | 54 ± 17 | 47.8 ± 16 | NS |
| Gender (M/F) | 13/9 | 18/6 | NS |
| APACHE II | 15 | 11.5 | NS |
| Ranson (48 h) | 5.5 | 6 | NS |
| CT | 1 (C); 8 (D); 13 (E) | 2 (C); 8 (D); 14 (E) | NS |
| Biliary pancreatitis | 14 (63.6 %) | 9 (37.5 %) | NS |
| Alcoholic pancreatitis | 5 (22.7 %) | 9 (37.5 %) | NS |
| Other pancreatitis | 3 (13.7 %) | 6 (25 %) | NS |

**Table 3** Complications (%)

| Type | Control | Somatostatin | p |
|---|---|---|---|
| Mechanical ventilation | 68.2 | 50 | NS |
| Renal failure | 45.5 | 41.7 | NS |
| Hemodialysis/ hemofiltration | 13.6 | 8.3 | NS |
| Sepsis | 90.9 | 87.5 | NS |
| Overall surgery | 86.4 | 45.8 | 0.005 |
| Surgery performed after 72 h in ICU | 63.6 | 37.5 | 0.07 |
| Pancreatic fistula | 9.1 | 8.3 | NS |
| Biliary fistula | 9.1 | 12.5 | NS |
| Intestinal fistula | 13.6 | 12.5 | NS |

**Somatostatin IV tambahan (selain terapi konvensional) diberikan kontinu selama 10 hari vs kontrol (terapi konvensional saja)**

- **Kebutuhan tindakan bedah keseluruhan lebih rendah pada kelompok somatostatin (45.8% vs 86.4% pada kontrol, p = 0.005)**
- **Operasi nekrosektomi lambat cenderung lebih jarang dengan somatostatin (37.5% vs 63.6%, p = 0.07)**

Planas M, Pérez A, Iglesia R, Porta I, Masclans JR, Bermejo B. Severe acute pancreatitis: treatment with somatostatin. Intensive Care Med. 1998;24(1):37-39. doi:10.1007/s001340050512

# *Meta-analysis of somatostatin, octreotide and gabexate mesilate in the therapy of acute pancreatitis*

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Table 4. Meta-analysis of clinical trials on somatostatin (SS), octreotide (OCT), and gabexate mesilate (FOY) in patients with severe acute pancreatitis

| | SS | | | OCT | | | FOY | | |
|---|---|---|---|---|---|---|---|---|---|
| | OR (95% CI) | Publ. bias assessment | NnT | OR (95% CI) | Publ. bias assessment | NnT | OR (95% CI) | Publ. bias assessment | NnT |
| Overall mortality | **A 0.36 (0.20–0.64)** | 14 | 11 | **0.57 (0.35–0.88)** | 3 | 14 | A 0.88 (0.56–1.39) | — | — |
| | **B 0.39 (0.18–0.86)** | 1 | 14 | 0.64 (0.38–1.09) | — | — | B 0.94 (0.55–1.62) | — | — |
| Early mortality | | | | | | | A 0.78 (0.35–1.76) | — | — |
| | | | | | | | B 1.03 (0.33–3.17) | — | — |
| Patients with complications | A 0.61 (0.33–1.14) | — | — | 0.97 (0.63–1.48) | — | — | **A 0.61 (0.41–0.91)** | 2 | 6 |
| | B 0.57 (0.26–1.23) | — | — | 0.97 (0.63–1.48) | — | — | **B 0.62 (0.41–0.93)** | 1 | 5 |
| Complication rate | — | — | — | — | — | — | **A 0.70 (0.56–0.88)** | 3 | 26 |
| | — | — | — | — | — | — | **B 0.70 (0.56–0.88)** | 3 | 26 |
| Need for surgery | — | — | — | — | — | — | **A 0.60 (0.39–0.92)** | 2 | 12 |
| | | | | | | | **B 0.75 (0.44–1.27)** | — | — |

A = all available trials.
B = only randomized trials.
Significant results are shown in bold characters.

# Efektifitas Somatostatin pada Pankreatitis Akut

**Meta-analisis Somatostatin pada pankreatitis akut**

- **Pada pankreatitis akut berat, somatostatin secara signifikan menurunkan mortalitas keseluruhan (OR 0,36; 95% CI 0,20–0,64; p = 0,001)**
- Octreotide juga menurunkan mortalitas (OR 0,57; p = 0,006), sedangkan gabexate tidak berpengaruh pada mortalitas
- Gabexate menurunkan insidensi komplikasi dan kebutuhan operasi (OR ~0,60; p ≈ 0,01)

Andriulli A, Leandro G, Clemente R, et al. Meta-analysis of somatostatin, octreotide and gabexate mesilate in the therapy of acute pancreatitis. Aliment Pharmacol Ther. 1998;12(3):237-245. doi:10.1046/j.1365-2036.1998.00295.x

GASTROENTEROLOGY

# Can somatostatin prevent post-ERCP pancreatitis? Results of a randomized controlled trial

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**Table 2** Endoscopic details of the procedure and frequency of pancreatitis

| | Placebo ($N$ = 122) | Som 1 ($N$ = 118) | Som 2 ($N$ = 116) |
|---|---|---|---|
| Sphincterotomy* | 104 | 106 | 108 |
| Yes† (%) | 12/104 (12) | 2/106 (2) | 2/108 (2) |
| Pancreatic duct injection** | 44 | 48 | 52 |
| Yes (%) | 9/44 (20) | 1/48 (2) | 1/52 (2) |
| Stone extraction*** | 58 | 58 | 60 |
| Yes (%) | 4/62 (6) | 1/58 (2) | 1/60 (2) |
| Stent placement**** | 20 | 22 | 24 |
| Yes (%) | 2 (10) | 0 (0) | 1/24 (4) |

Comparing the placebo group with both som 1 and som 2 groups the differences were: *$P < 0.01$; **$P < 0.05$; ***NS (not significant); ****NS. Between the som 1 and som 2 groups no statistically significant differences existed. † Cases with acute pancreatitis as a complication of the procedure.

**Table 3** Amylase levels and percentage of hyperamylasemia before and after endoscopic retrogade cholangiopancreatography (ERCP) in all groups

| Amylase levels* | Placebo | Som 1*** | Som 2 |
|---|---|---|---|
| Baseline (U/L) ± SD | 92 ± 13 | 87 ± 17 | 91 ± 20 |
| After 5 h (U/L) ± SD | 393 ± 35 | 204 ± 48 | 207 ± 33 |
| After 24 h (U/L) ± SD | 308 ± 45 | 137 ± 54 | 148 ± 61 |
| Hyperamylasemia** | | | |
| After 5 h (%) | 66/122 (54) | 44/118 (37) | 46/116 (40) |
| After 24 h (%) | 52/122 (47) | 32/118 (27) | 31/116 (27) |

*Differences between placebo group and som 1 and som 2 groups regarding amylase levels at 5 and 24 h after ERCP were statistically significant ($P < 0.05$); **differences in rates of hyperamylasemia between the placebo group and the som 1 and som 2 groups at 5 and 24 h after ERCP were statistically significant ($P < 0.05$); ***there were no statistically significant differences between the som 1 and som 2 groups regarding either amylase levels or the frequency of hyperamylasemia after ERCP.

# Efektifitas Somatostatin pada Pankreatitis Akut

Grup 1 (**bolus somatostatin**): Dosis 4 mg/kg berat badan somatostatin melalui suntikan intravena bolus, ditambah infus somatostatin 500 mL normal saline selama 12 jam

Grup 2 (**infus somatostatin**): Dosis 3 mg somatostatin dalam 500 mL normal saline, diberikan melalui infus selama 12 jam

Grup kontrol (plasebo): Infus normal saline 500 mL selama 12 jam dan suntikan intravena normal saline saat kanulasi papilla

Durasi pemberian: 12 jam, mulai 1 jam sebelum prosedur ERCP

Hasil:

Insiden pankreatitis pasca-ERCP:

- **Grup somatostatin (bolus dan infus): 1.7%**
- Grup plasebo: 9.8%, P-value: < 0.05, hasil signifikan

**Tingkat amilase serum pada 5 dan 24 jam setelah ERCP lebih rendah pada grup somatostatin** dibandingkan grup plasebo (P < 0.05)

Tidak ada perbedaan signifikan antara kedua grup somatostatin (bolus vs infus)

Andriulli A, Leandro G, Clemente R, et al. Meta-analysis of somatostatin, octreotide and gabexate mesilate in the therapy of acute pancreatitis. Aliment Pharmacol Ther. 1998;12(3):237-245. doi:10.1046/j.1365-2036.1998.00295.x

# The Prophylactic Effect of Somatostatin on Post-Therapeutic Endoscopic Retrograde Cholangiopancreatography Pancreatitis

## *A Randomized, Multicenter Controlled Trial*

*Kyu Taek Lee, MD, PhD,* Don Haeng Lee, MD, PhD,† and Byung Moo Yoo, MD, PhD‡*

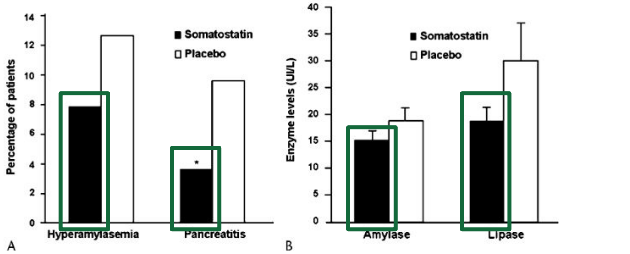


**FIGURE 1.** A, Percentage of patients who had hyperamylasemia and pancreatitis 24 h after their ERCP procedure. *$P$ = 0.02. B, Mean enzyme levels observed 24 h after ERCP procedures (error bars show SEM).

**TABLE 3.** Pancreatic Enzyme Levels and Pancreatitis at 24 h After ERCP

| | Somatostatin | Placebo | P |
|---|---|---|---|
| Amylase, mean (SD), IU/L | 151 (268) | 185 (404) | 0.32 |
| Lipase, mean (SD), IU/L | 186 (423) | 298 (1048) | 0.17 |
| Hyperamylasemia | 15/193 (7.8%) | 25/198 (12.6%) | 0.13 |
| Pancreatitis | 7/193 (3.6%) | 19/198 (9.6%) | 0.02 |

**TABLE 2.** Incidence of Pancreatitis for Each Endoscopic Procedure

| Procedure | Somatostatin | Placebo | P |
|---|---|---|---|
| EST | 104 (54.5%) | 106 (53.5%) | |
| [Pancreatitis] | [5/104 (4.8%)] | [12/106 (11.3%)] | 0.08 |
| Stone removal | 96 (50.3%) | 118 (59.6%) | |
| [Pancreatitis] | [2/96 (2.1%)] | [11/118 (9.3%)] | 0.04 |
| Stenting* | 98 (51.3%) | 75 (37.9%) | |
| [Pancreatitis] | [4/98 (4.1%)] | [8/75 (10.7%)] | 0.13 |
| EPBD | 47 (24.6%) | 45 (22.7%) | |
| [Pancreatitis] | [1/47 (2.1%)] | [7/45 (15.6%)] | 0.03 |
| EPT | 1 (0.5%) | 0 (0%) | |
| [Pancreatitis] | [0] | [0] | |
| Other | 2 (1.0%) | 0 (0%) | |
| [Pancreatitis] | [0] | [0] | |

*This was done exclusively for bile duct stenting.

3 mg somatostatin dalam 500 mL saline, diberikan melalui infus selama 12 jam, dimulai 30 menit sebelum prosedur ERCP vs plasebo (500 mL saline yang diinfus selama 12 jam, dimulai 30 menit sebelum prosedur ERCP)

Insiden pankreatitis:

- Grup somatostatin: 3.6% (7/193 pasien)
- Grup plasebo: 9.6% (19/198 pasien), P-value: 0.02, hasil signifikan

Insiden hyperamylasemia:

- Grup somatostatin: 7.8% (15/193 pasien)
- Grup plasebo: 12.6% (25/198 pasien)

Prosedur spesifik (seperti pengangkatan batu dan EPBD) menunjukkan penurunan signifikan dalam insiden pankreatitis pada grup somatostatin (P = 0.04 untuk pengangkatan batu dan P = 0.03 untuk EPBD)

Andriulli A, Leandro G, Clemente R, et al. Meta-analysis of somatostatin, octreotide and gabexate mesilate in the therapy of acute pancreatitis. Aliment Pharmacol Ther. 1998;12(3):237-245. doi:10.1046/j.1365-2036.1998.00295.x

# Take Home Messages

| |
|---|
| 30–40 % sirosis kompensasi → varices esofagus<br>Mortalitas BEV 15–25 % dalam 6 minggu<br>Somato bolus 250 μg IV, lanjut 250–500 μg/jam 48 jam<br>Capai hemostasis 81 %, turunkan mortalitas pada BEV<br>BEV: EVL + somatostatin ≤12 jam meningkatkan hemostasis<br>somatostatin lebih aman daripada vasopresin & terlipressin |
| Insidensi Pankreatitis akut 79,8/100.000<br>Somatostatin 250 μg/jam menurunkan mortalitas & lama rawat pankreatitis berat<br>Insidens komplikasi lebih rendah dibandingkan plasebo |

# Terima Kasih